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Computer Vision System for Medication Intake Monitoring Through Multi-Modal Fusion

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Abstract

The present study proposes the development and validation of an end-to-end computer vision system for objective medication intake monitoring through multi-modal fusion of YOLOv8 object detection, MediaPipe pose estimation, and ResNet-18 action classification technologies. The system utilises a vision-only, single-camera approach with privacy-preserving architecture that processes skeletal keypoints while discarding identifiable images. This is integrated through a transparent finite-state machine controller that enforces temporal event sequences and timing constraints. The methodology addresses critical limitations in existing adherence monitoring approaches by providing objective, non-invasive behavioural verification without modifying existing medication routines or requiring invasive sensor technologies. The research establishes a comprehensive framework encompassing the construction of datasets with quality controls, the integration of multimodal components, temporal fusion strategies, and clinical interpretability requirements. This offers a scalable approach to medication adherence monitoring suitable for deployment in both home and clinical environments while maintaining computational efficiency and regulatory compliance.

Keywords: Medication adherence monitoring; Computer vision; Multi-modal fusion; Object detection; Pose estimation; Action recognition; YOLO; MediaPipe; ResNet; Healthcare technology; Behavioral monitoring; Machine learning; Deep learning

1 Introduction

Non-adherence to medication regimens constitutes a persistent challenge within the domains of clinical care and home health, exerting a substantial impact on outcomes and associated costs. The World Health Organization has long emphasised that enhancing adherence can exert a more substantial influence on population health than advancements in specific medical treatments (*World Health Organization, 2003*). The development of a camera-based monitor capable of verifying not only the occurrence of a dose but also the method of administration (e.g., oral ingestion followed by drinking) would provide clinicians with objective evidence of adherence. In addition, this technology could offer families a practical method of supporting elders living independently. In contradistinction to self-reports or uncomplicated container logs, a computer-vision system is capable of observing the interaction between the face, hand and container, and of discriminating between partial or erroneous behaviours and a complete intake episode (*Bilodeau & Ammouri, 2011*).

Alternative adherence technologies, such as electronic pill caps “Medication Event Monitoring System”(MEMS) and pharmacy refill metrics, offer value but come with workflow friction and measurement biases. Systematic comparisons demonstrate that electronic event caps measure a different construct to self-report and claims data, and each method has distinct failure modes in practice (*El Alili et al., 2016*). At the opposite end of the spectrum, ingestible sensors can confirm ingestion with a high degree of certainty; however, this is achieved at the cost of additional hardware, integration burden (given its invasive nature) and acceptability concerns in everyday use (*Hafezi et al., 2015*). Vision offers a complementary approach: a single

commodity camera can observe the relevant interactions without modifying the medication itself, provided the subject is cooperative in a home or clinic context.

Early vision systems for medication intake relied on skin-colour cues, face/hand tracking, and rule engines (e.g. Petri nets) that fired when proximity and occlusion patterns persisted, demonstrating feasibility under controlled conditions (*Bilodeau & Ammouri, 2011*). Recent advancements in real-time object detectors and keypoint trackers have prompted the development of a more robust formulation that maintains clinician-friendly interpretability. The foundational You Only Look Once (YOLO) architecture revolutionised real-time object detection through unified, single-stage processing (*Redmon et al., 2016*), while subsequent refinements have optimised performance for challenging detection scenarios (*Lin et al., 2017*). For targets of diminutive size and substandard texture, such as tablets and capsules, one-stage detectors in the YOLO family offer a favourable speed-accuracy trade-off, particularly when compared to two-stage alternatives such as Fast R-CNN (*Girshick, 2015*) and region proposal networks (*Ren et al., 2015*). In a pharmacy pill-identification benchmark, YOLOv3 achieved real-time throughput while maintaining competitive detection quality (*Tan et al., 2021*). In the context of human motion cues, the utilisation of on-device pose and hand landmarking with lightweight networks, such as BlazePose for a 33-keypoint body model or MediaPipe Hands for 21-point hand skeletons, ensures the stability of geometry in RGB video at interactive frame rates. This, in turn, facilitates reliable reasoning about wrist–mouth proximity without the requirement of storing identifiable imagery (*Bazarevsky et al., 2020; Zhang et al., 2020*). Finally, compact convolutional backbones, such as ResNet-18 (and/or ResNet34/50/101/152), have been shown to provide efficient, transferable visual features for frame-level action classification within the resource constraints typical of edge deployments (*He et al., 2016*).

Guided by this landscape, the project provides a framework for the analysis of pill intake as a series of observable events captured in a monocular RGB video. This framework captures four distinct phases: container manipulation (opening/closing), pill appearance in hand, hand-to-mouth contact, and subsequent drinking. The technical contribution is constituted by an end-to-end pipeline that: constructs a custom, task-specific dataset, detects pills and bottles with a fine-tuned YOLO model and extracts upper-body and hand landmarks with MediaPipe to supply a scale-normalized wrist–mouth proximity signal. The fourth step involves the classification of intake-related phases using a ResNet-18 image classifier. The fifth step is the fusion of these streams with a transparent, finite-state controller that operates in real time. The controller emits auditable artefacts (timelines, overlaid frames, JSON event logs) for clinical review. Skeleton-centric reasoning is also consistent with privacy-preserving practices in healthcare activity monitoring, where landmark sequences can facilitate recognition while minimising exposure of identifiable video content (*Sardari et al., 2023*).

Building on this motivation, the following sections set out the plan of work, outlining the guiding research questions and the aims and objectives that shape the methodology.

1.1 Research Questions

This study addresses four fundamental questions arising from the operational constraints of home and clinic monitoring:

Can a monocular, real-time vision system reliably detect pill-intake events in unseen videos under subject-wise splits, achieving clinically useful accuracy for medication adherence monitoring?

Can the system verify that intake was performed correctly by recovering the canonical sequence of medication administration with sufficient precision and recall to support clinical and family decision-making?

Which modelling and data-handling choices most improve reliability across subjects, cameras, and lighting conditions, including detection thresholds, pose parameters, temporal persistence requirements, and dataset curation strategies?

Is the end-to-end system sufficiently robust, efficient, and privacy-preserving for real-world deployment, and what human oversight mechanisms are required to ensure safe clinical implementation?

1.2 Aims and Objectives

Aim.

The objective of this project is to develop and validate an end-to-end, real-time system that infers medication intake from monocular RGB video through multi-modal computer vision approaches. The system has been designed to achieve clinically useful event-level performance while maintaining privacy preservation and computational efficiency suitable for deployment in home and clinical environments.

Objective 1: Dataset Development The establishment of a domain-specific dataset encompassing medication intake behaviours is imperative, accompanied by the implementation of comprehensive quality controls, ethical safeguards, and reproducible annotation protocols.

Objective 2: Small Object Detection The development of robust detection capabilities for pharmaceutical objects under realistic environmental conditions is imperative, with a focus on evaluating trade-offs between accuracy and computational efficiency.

Objective 3: Privacy-Preserving Pose Analysis It is essential to implement reliable pose and hand geometry extraction that maintains behavioural monitoring capability while eliminating identifiable visual content.

Objective 4: Temporal Action Recognition The creation of lightweight action classification systems capable of recognising medication-related behaviours with appropriate sensitivity to clinical requirements is essential.

Objective 5: Multi-Modal Integration The objective is to achieve real-time fusion of complementary sensing modalities through interpretable decision-making processes that support clinical audit and review.

Objective 6: Clinical Validation Framework The establishment of transparent evaluation protocols is imperative for the quantification of system performance, robustness, and deployment readiness under realistic operational conditions.

The remainder of this paper proceeds as follows. The remainder of this paper proceeds as follows. The second section provides a comprehensive overview of the existing literature on the subject, as well as detailing the construction and packaging of the dataset. Sections III and IV present the pill-detection module (YOLO) and the pose/hand-landmark pipeline (MediaPipe) in conjunction with the action classifier (ResNet-18). The fifth section of this text provides a detailed exposition on the fusion controller for final intake prediction. Sections VI and VII present the results and evaluation, incorporating ablations, latency, and error analysis. Section VIII concludes with limitations and directions for improvement.

2 Literature Review

The issue of medication non-adherence has been identified as one of the most significant challenges facing healthcare systems today. It is estimated that this problem affects between 50 and 70 per cent of patients on a global scale, and it plays a crucial role in the occurrence of preventable hospital admissions, disease progression, and mortality (*World Health Organization, 2003*). Despite the World Health Organization's long-standing emphasis on the potential of enhancing adherence to exert a more substantial influence on population health than advancements in specific medical treatments (*World Health Organization, 2003*), the technological solutions developed to address this challenge have remained fragmented, invasive, or inadequately validated for real-world deployment. This literature review examines the evolution of medication adherence monitoring technologies, with particular focus on computer vision approaches that can provide objective, non-invasive verification of intake behaviours without compromising patient privacy or workflow integration.

The central thesis of this work, that multi-modal computer vision can provide clinically useful adherence monitoring through the fusion of object detection, pose estimation, and action recognition, emerges from critical limitations observed in existing approaches. As is well documented, traditional approaches to measurement are often subject to various biases (*El Alili et al., 2016*) whilst emerging technologies often prioritise technical complexity over practical deployability. This review therefore adopts a critical lens, evaluating not only the technical achievements of prior work but also their translational potential and alignment with the real-world constraints of home and clinical monitoring environments.

2.1 Medication Adherence Monitoring: From Traditional Methods to Technological Solutions

2.1.1 The Fundamental Challenge of Adherence Measurement

The accurate measurement and monitoring of patient medication adherence is a global challenge due to the absence of gold standard methods for adherence measurement (*Alfian et al., 2022*). This fundamental observation underlies decades of research attempting to develop reliable, scalable adherence monitoring solutions. The absence of a "gold standard" reflects underlying methodological challenges; adherence is not merely a binary state, but a complex behavioural phenomenon influenced by cognitive, social, economic, and clinical factors that

traditional measurement approaches struggle to capture comprehensively. The conventional triad of adherence measurement, self-reporting, pill counting and pharmacy refill tracking, each demonstrates systematic biases that curtail their clinical utility. Although self-reporting is scalable and non-invasive, it is subject to recall bias, social desirability effects, and a fundamental disconnect between patient perception and objective behaviour (*Ahuja et al., 2021*). Pill counting provides an objective evidential basis, but it is not possible to distinguish between pills removed from containers and pills actually consumed. This creates opportunities for intentional deception or accidental miscounting. The increasing availability of pharmacy refill data through electronic health records measures prescription acquisition rather than consumption, and fails to capture complex behaviours such as dose timing, drug holidays, or medication sharing. The literature consistently emphasises that these measurement limitations are not just technical issues, but rather reflect a deeper misalignment between the information required for clinical decision-making and the behavioural aspects of medication use (*El Alili et al., 2016; Vrijens et al., 2012*). In order to adjust dosing, identify contraindications and target interventions, clinicians require timely, actionable insights into adherence patterns (*Krousel-Wood et al., 2004*). However, the majority of methods yield retrospective, aggregate indicators that obscure temporal dynamics and context (*World Health Organization, 2003; El Alili et al., 2016*). This critical appraisal of previous studies has led to the development of computer vision approaches that can capture the behavioural sequence of medication intake with temporal precision while maintaining the objectivity required for clinical applications.

2.1.2 Electronic Monitoring Systems: Promises and Limitations

Electronic Medication Event Monitoring Systems (MEMS) represent the most widely adopted technological approach to adherence monitoring, utilising pressure-sensitive caps that record container opening events with timestamps (*El Alili et al., 2016*). Whilst MEMS systems provide objective, time-stamped data that supports both clinical research and patient care, systematic comparisons reveal critical limitations that motivate the exploration of alternative approaches. A scoping review of studies comparing the medication event monitoring system (MEMS) with alternative methods for measuring medication adherence demonstrates that electronic event caps measure a different construct to self-report and claims data, and each method has distinct failure modes in practice (*El Alili et al., 2016*). This fundamental finding calls into question the assumption that technological sophistication necessarily translates to measurement validity. MEMS systems capture "container access" rather than "medication consumption," creating systematic opportunities for measurement error when patients remove pills without taking them, transfer medications to alternative containers, or engage in dose dumping behaviours prior to clinical appointments. The workflow friction introduced by MEMS systems presents additional barriers to widespread adoption. Patients must be mindful of the necessity to utilise specialised containers, manage multiple capsules for complex regimens, and tolerate the inconvenience of bulky, expensive monitoring devices. These requirements have been shown to create selection biases in research studies (*Byerly et al., 2007; Velligan et al., 2006*) and limit real-world scalability, particularly among elderly populations who may struggle with technology adoption (*Granger et al., 2005; Russell et al., 2006*) or patients with complex polypharmacy regimens that require multiple monitoring devices (*Conn et al., 2009; Krousel-Wood et al., 2004*). The analysis suggests that the limitations of MEMS systems stem from their reliance on proxy measurements rather than direct behavioural observation. Container opening events provide indirect evidence of intake intentions; however, they cannot verify that the retrieved medication was consumed according to prescribed

instructions. This creates a fundamental gap between the construct of interest (medication consumption behaviour) and the measured outcome (container access patterns) that no amount of data processing or statistical modelling can fully bridge. Computer vision approaches offer a compelling alternative by enabling direct observation of the behavioural sequence that constitutes medication intake, potentially providing higher construct validity while maintaining objective measurement properties.

2.1.3 Ingestible Sensors: The Technical Ceiling

The theoretical upper limit of adherence monitoring precision is represented by ingestible sensors, which utilise FDA-approved digital pills that combine medications with embedded sensors that activate upon contact with gastric fluid (*Hafezi et al., 2015*). The utilisation of artificial intelligence (AI) tools within the domain of healthcare is witnessing a marked increase, with these tools being employed for a variety of purposes, including the facilitation of patient adherence to drug regimens (*Milosevic et al., 2021*). However, the deployment of ingestible sensors, while offering the potential for verification certainty, is accompanied by significant implementation complexity and patient acceptability concerns. The clinical trials of ingestible sensor systems have demonstrated technical feasibility and measurement precision that exceeds all alternative monitoring approaches (*Hafezi et al., 2015*). The system provides definitive confirmation of pill ingestion with temporal precision, thereby enabling real-time adherence tracking and immediate intervention opportunities. However, the practical implementation requirements, namely, specialised medication formulations, body-worn sensor patches, smartphone applications for data transmission, and clinical workflow integration, create substantial barriers to routine deployment. The acceptability challenges associated with ingestible sensors are of particular pertinence within the context of our research. Patient surveys consistently identify concerns about "being tracked" or having technology "inside their body" as primary barriers to adoption (*Hafezi et al., 2015*). These concerns are particularly pronounced among elderly populations and patients with mental health conditions, who are precisely the demographics most likely to benefit from adherence monitoring interventions. Moreover, the regulatory intricacies inherent in the combination of pharmaceutical products with electronic devices give rise to approval pathways that are characterised by a considerable delay in clinical availability and an increase in costs. From a systems design perspective, ingestible sensors represent a sophisticated yet overly complex solution to the adherence monitoring challenge. While these devices achieve maximal measurement precision, they do so by fundamentally altering the medication experience in ways that may paradoxically reduce adherence through increased complexity and patient resistance. This observation lends further support to the notion that effective adherence monitoring requires solutions that integrate seamlessly into existing medication routines, as opposed to replacing them with technologically sophisticated alternatives.

The present study highlights a critical design principle adopted as computer vision approach, namely that monitoring solutions should observe rather than alter medication behaviours, thereby maintaining the naturalness of existing routines while providing objective verification capabilities. This principle is evidenced by the significant gap between ingestible sensor capabilities and practical deployment constraints.

2.2 Computer Vision in Healthcare: Technical Foundations and Clinical Applications

2.2.1 The Evolution of Medical Computer Vision

The application of computer vision to healthcare has undergone rapid evolution, driven by advances in deep learning architectures (*Szegedy et al., 2015; Krizhevsky et al., 2017*), increased computational accessibility, and growing availability of medical imaging datasets (*Fei-Fei et al., 2007; Deng et al., 2009*). The transition from traditional computer vision approaches to deep learning-based systems has been particularly transformative in medical applications, where the availability of large-scale annotated datasets such as ImageNet (*Deng et al., 2009*) has enabled effective transfer learning to specialised healthcare domains. However, the translation of computer vision successes from consumer applications to clinical environments presents unique challenges that distinguish medical applications from general-purpose vision tasks.

The challenges specific to medical computer vision applications include stringent accuracy requirements that exceed those of typical consumer applications (*Rajpurkar et al., 2022; Liu et al., 2019*), the need for interpretable decision-making processes that support clinical reasoning (*Holzinger et al., 2017; Rudin, 2019*), and regulatory frameworks that demand extensive validation and risk assessment (*FDA, 2021; Muehlematter et al., 2021*). Furthermore, medical environments present unique technical challenges related to variable lighting conditions, diverse patient populations, privacy constraints (*Price & Cohen, 2019; Kaissis et al., 2020*), and the need for real-time processing in resource-constrained settings (*Litjens et al., 2017; Esteva et al., 2017*). This observation reflects a broader pattern in the field of medical computer vision: while there has been rapid advancement in technical capabilities, clinical validation and real-world deployment have lagged significantly behind laboratory demonstrations (*Rasmussen et al., 2023; Kelly et al., 2019*).

The present approach to medication intake monitoring builds upon established computer vision foundations, while addressing the specific constraints of healthcare applications. The selection of YOLO architectures for object detection, MediaPipe for pose estimation, and ResNet for action classification is indicative not only of their technical capabilities but also of their suitability for clinical deployment, taking into consideration factors such as computational requirements, interpretability, and robustness to real-world variation.

2.2.2 YOLO Architectures in Medical Object Detection

The You Only Look Once (YOLO) family of object detection algorithms was first introduced by Redmon et al. for unified real-time object detection (*Redmon et al., 2016*). It has since gained significant traction in medical applications thanks to its optimal balance between accuracy and computational efficiency (*Rahman et al., 2024*). The single-stage detection paradigm eliminates the computational overhead of the region proposal networks employed by two-stage detectors (*Girshick, 2015; Ren et al., 2015*), and innovations in focal loss address the class imbalance issues commonly encountered in medical detection tasks (*Lin et al., 2017*). Rahman et al. (2024) conducted a thorough and methodical review of the literature on the use of YOLO for medical object detection between 2018 and 2023. This review identified 124

studies using YOLO for various tasks, such as lesion detection, skin lesion classification, retinal abnormality identification, cardiac abnormality detection, brain tumour segmentation, and personal protective equipment monitoring. The extensive adoption of YOLO documented by Rahman et al. validates its suitability for medical computer vision and highlights its diverse potential applications. The architectural advantages of YOLO for medical applications extend beyond computational efficiency to include practical deployment considerations. The employment of single-stage detectors serves to eliminate the complexity inherent in region proposal networks, thereby reducing the number of hyperparameters requiring tuning and concomitantly simplifying the training pipeline. This architectural simplicity is reflected in the fact that it is more predictable across a range of medical imaging conditions and is less sensitive to optimisations specific to a given dataset, which may not generalise to new clinical environments. With regard to the specific task of pill detection, the manner in which YOLO handles small objects gives rise to both opportunities and challenges. It is noteworthy that pharmaceutical tablets and capsules typically occupy only a small fraction of the captured frames, thereby creating the so-called "small object detection" challenge. This challenge has been the subject of extensive study in the domain of computer vision literature (*Tan et al., 2021*). A comparison of RetinaNet, SSD, and YOLO v3 for real-time pill identification demonstrates that YOLO architectures can achieve competitive detection performance while maintaining real-time processing requirements (*Tan et al., 2021*). However, an analysis of this work reveals significant limitations in the evaluation methodology that affect interpretation of results for clinical applications.

Alternative object detection approaches comprise two-stage detectors, such as Faster R-CNN (*Ren et al., 2015*) and Mask R-CNN (*He et al., 2017*), which employ separate region proposal and classification networks. The superior accuracy of Faster R-CNN is achieved through its Region Proposal Network (RPN), which generates high-quality object proposals, followed by a dedicated classification head that refines detection results. This two-stage approach has been shown to achieve higher mean Average Precision (mAP) scores, particularly in complex scenes involving overlapping objects or challenging backgrounds. However, the sequential processing architecture requires substantially greater computational resources, with inference times often exceeding 100 milliseconds per frame on standard hardware, making real-time applications challenging. Mask R-CNN extends Faster R-CNN by incorporating instance segmentation capabilities through a parallel mask prediction branch, thereby enabling pixel-level object delineation in conjunction with bounding box detection. While this does provide richer spatial information that could theoretically benefit pill localisation accuracy, the additional computational overhead and complexity result in its unsuitability for resource-constrained healthcare deployments where simplicity and reliability are paramount.

Single Shot Detector (SSD) architectures (*Liu et al., 2016*) provide another single-stage alternative that performs detection in a single forward pass through the network. SSD employs multiple feature maps at different scales to detect objects of varying sizes, which theoretically benefits small object detection. However, empirical comparisons consistently demonstrate that SSD variants generally achieve lower accuracy than YOLO architectures for small objects like pharmaceutical tablets, particularly in cluttered backgrounds typical of home environments (*Tan et al., 2021*). The adoption of an SSD approach necessitates a more intricate anchor box configuration and hyperparameter tuning, thereby augmenting deployment complexity. The selection of YOLO reflects the optimal balance between detection accuracy and computational efficiency required for real-time healthcare monitoring applications. The single-stage architecture of YOLO eliminates the complexity of region proposal networks while maintaining competitive accuracy for small object detection. The unified detection framework

has been shown to simplify training and deployment pipelines (*Gholamy et al., 2018*), thereby reducing the potential for configuration errors that could compromise system reliability in clinical settings. Furthermore, the consistent inference timing of YOLO enables predictable system response, which is essential for real-time monitoring applications where bounded latency is critical for timely intervention (*Ultralytics, 2023*).

The Tan et al. study (2021), while pioneering in directly addressing pill detection, employed controlled imaging conditions and limited pill varieties that may not reflect the complexity of real-world medication monitoring scenarios. The photographic documentation of the pills was conducted in controlled settings, with a focus on uniform backgrounds and consistent lighting conditions. This approach served to mitigate the numerous visual challenges that are often present in domestic environments, including cluttered backgrounds, variable lighting conditions, and partial occlusions by hands or containers. Furthermore, the study concentrated on pill identification (the differentiation between various medication types) as opposed to pill detection in the context of intake behaviours, thus overlooking the temporal and contextual complexities that characterise medication administration sequences. The present approach is founded on the technical foundations established by Tan et al., whilst addressing several critical limitations. Firstly, the focus is on binary pill detection rather than multi-class identification, thus reducing complexity while maintaining clinical relevance for adherence monitoring applications. Secondly, the construction of the dataset incorporates realistic environmental conditions, hand occlusions and motion blur, which are characteristic of actual medication intake scenarios. Thirdly, the pill detection has been integrated within a more extensive behavioural monitoring framework that captures the temporal sequence of intake actions, as opposed to treating pill detection as an isolated computer vision task. The transition from a system of controlled pill identification to the implementation of behavioural intake monitoring signifies a paradigm shift in the formulation of problems. While the identification of specific medications from static images provides value for medication verification and inventory management, adherence monitoring requires understanding medication intake as a temporal behavioural sequence that includes container manipulation, pill retrieval, hand-to-mouth contact, and consumption verification. The present study adopts a systems perspective that differentiates its approach from previous research on pill detection and aligns with the broader clinical requirements of adherence monitoring applications.

2.2.3 Recent Advances in Pharmaceutical Computer Vision

Building upon foundational work in pill detection, recent research has begun to address more complex pharmaceutical monitoring challenges that are better aligned with clinical requirements. Research on the detection of toxic and narcotic medication using rotated object detectors has demonstrated advances in the handling of arbitrary pill orientations and complex pharmaceutical packaging (*Cao et al., 2017*). The present work addresses a critical limitation of early pill detection systems that assumed canonical orientations and simple geometric presentations.

The management of rotational variance in pill detection represents a significant technical advancement with substantial clinical implications. In real-world medication intake scenarios, pills are often found in arbitrary orientations because patients handle them, either by removing them from containers or by placing them on surfaces prior to consumption. Traditional object detection approaches based on axis-aligned bounding boxes are ineffective under these

conditions, resulting in missed detections or false positives that compromise monitoring reliability. However, an analysis of the existing literature on rotated detection reveals significant discrepancies between the technology's capabilities and the clinical requirements for its use (Yang *et al.*, 2021; Zhou *et al.*, 2017; Ding *et al.*, 2019). The majority of rotated detection approaches focus on isolated object recognition rather than contextual behavioural understanding (Han *et al.*, 2022; Xu *et al.*, 2021). For example, Yang *et al.* (2021) demonstrate improved detection of rotated objects in aerial imagery, and Zhou *et al.* (2017) make advances in oriented text detection. However, these methods focus primarily on geometric accuracy rather than temporal or behavioural context. Similarly, recent work on rotated bounding box regression emphasises precision in object localisation, but lacks integration with sequential action understanding. The clinical value of detecting a pill in an arbitrary orientation is limited in the absence of temporal context relating to the patient's intentions and behaviours. The fusion approach combines object detection with pose estimation and action classification, thereby providing a more comprehensive behavioural understanding than isolated object recognition alone.

The publication of clinical validation studies of pill detection systems in high-impact medical journals has indicated a growing acceptance of computer vision approaches in healthcare applications. Research published in the Journal of Medical Internet Research demonstrates a deep learning-based system for reducing medication errors by accurately identifying prescription pills (Tan *et al.*, 2021). This work signifies a significant advancement towards clinical translation by shifting the focus from purely technical computer vision challenges to the realm of medication safety. The clinical focus on error reduction rather than adherence monitoring is indicative of significant priorities in healthcare applications of computer vision. Medication errors, encompassing the administration of incorrect medication, dosage, or timing, pose imminent safety concerns that substantiate technological intervention. Adherence monitoring, while of significance for long-term health outcomes, encompasses a more intricate web of behavioural and social factors that resist purely technological solutions. The present approach is predicated on the recognition of this distinction by focusing on objective behavioural verification rather than attempting to solve the broader behavioural and social dimensions of adherence through technology alone.

2.3 Human Pose Estimation and Behavioural Monitoring in Healthcare

2.3.1 MediaPipe and Real-Time Pose Estimation

The advent of production-ready pose estimation systems through Google's MediaPipe framework has made human pose estimation, which was previously limited to specialised labs, more accessible to a wider range of users (Bazarevsky *et al.*, 2020; Zhang *et al.*, 2020). The MediaPipe Pose Landmarker detects landmarks on the human body in images or videos, locating key points to analyse posture and categorise movements (Google AI for Developers, 2024). This accessibility has enabled the rapid deployment of pose-based healthcare applications, though there are caveats regarding accuracy, robustness and clinical validation. Lightweight, deployment-oriented models such as BlazePose can achieve real-time speeds on mobile and web runtimes, but they sacrifice accuracy in situations involving occlusions and non-standard viewpoints, which are common in medication monitoring (Bazarevsky *et al.*,

2020). Alternative frameworks include OpenPose (Cao *et al.*, 2017), which pioneered real-time multi-person estimation using Part Affinity Fields (PAFs). OpenPose offers high accuracy with ~25 body keypoints using a bottom-up pipeline that first detects all keypoints and then associates them with individuals. However, this comes at the cost of substantial computation (often >50 ms/frame on a single GPU), which limits its deployment in home healthcare settings. Subsequent efforts in efficient architectures prioritise deployment efficiency over raw accuracy (Bazarevsky *et al.*, 2020; Fang *et al.*, 2017; Sun *et al.*, 2019). Although AlphaPose (Fang *et al.*, 2017) and HRNet (Sun *et al.*, 2019) deliver state-of-the-art accuracy via sophisticated designs, such as SSTN/P-NMS and high-resolution multi-scale fusion, they typically exceed the computing and memory capabilities of standard edge hardware. MediaPipe’s production-ready architecture strikes a pragmatic balance between accuracy and efficiency. BlazePose employs a two-stage pipeline comprising a lightweight person detector followed by a 33-keypoint regressor. This delivers sufficient accuracy for medication-monitoring tasks while maintaining real-time performance on mobile and edge devices. Cross-platform optimisations and integrated hand tracking reduce system complexity and failure modes (Bazarevsky *et al.*, 2020; Zhang *et al.*, 2020). Research on privacy-preserving in-bed pose monitoring shows that processing only the skeleton can retain clinical utility while protecting patient privacy (Li *et al.*, 2023). Processing streams to extract skeletal keypoints and then discarding RGB imagery supports behavioural monitoring without retaining identifiable data, thus aligning with healthcare data protection requirements.

In this work, the privacy-preserving capabilities of MediaPipe are a core design principle. Hand-to-mouth detection requires precise spatial localisation and temporal consistency, placing a high demand on landmark stability during subtle, occlusion-prone hand–face interactions. These limitations are overcome by using multi-modal fusion to combine pose detection with object detection and action classification. This provides redundant evidence streams and enables more reliable behavioural monitoring than any single component alone.

2.3.2 Privacy-Preserving Healthcare Monitoring

The privacy implications of video-based healthcare monitoring pose significant challenges to its clinical adoption, extending beyond technical capabilities to include legal, ethical and social considerations (Al-Taei *et al.*, 2018). Regulations pertaining to the protection of healthcare data, including the Health Insurance Portability and Accountability Act (HIPAA) in the United States, the General Data Protection Regulation (GDPR) in Europe and the UK Data Protection Act 2018, which incorporates GDPR principles following Brexit, establish rigorous requirements for minimising patient data and implementing security controls. These requirements directly influence the design of computer vision systems (Randhawa *et al.*, 2022). The skeleton-based approach to privacy preservation addresses these regulatory requirements while maintaining sufficient behavioural information for clinical monitoring applications. Research on intelligent, privacy-preserving medication adherence systems has demonstrated methods of protecting patient data while maintaining monitoring effectiveness (Al-Tawil *et al.*, 2023). However, the practical implementation of privacy-preserving monitoring involves complex trade-offs between privacy protection, monitoring accuracy, and clinical utility that require careful engineering and validation.

The skeleton-based approach to privacy preservation offers theoretical advantages but requires empirical validation to ensure that clinical utility is maintained when visual information is

discarded. Skeletal keypoints have been shown to preserve spatial relationships and temporal dynamics that facilitate behavioural analysis (*Randhawa et al., 2022*). This is achieved by eliminating facial features, clothing details and environmental context that could potentially enable patient identification. However, the information loss associated with skeleton-only processing may compromise the ability to detect subtle behavioural cues or environmental factors that influence adherence behaviours.

The present study directly addresses this privacy-utility trade-off through systematic evaluation of skeleton-only monitoring capabilities. The present study aims to compare the performance of behavioural detection using full video streams versus skeleton-only representations. This will allow the quantification of the accuracy costs associated with privacy preservation. Furthermore, it will enable the identification of optimal operating points that balance privacy protection with clinical utility. This empirical approach to privacy-utility optimisation provides practical guidance for healthcare system deployment while advancing the theoretical understanding of privacy-preserving behavioural monitoring. The regulatory and legal landscape surrounding video-based healthcare monitoring introduces further intricacies to privacy considerations. Healthcare data protection regulations, such as the Health Insurance Portability and Accountability Act (HIPAA) in the United States, the General Data Protection Regulation (GDPR) in Europe and the UK Data Protection Act 2018, which incorporates GDPR provisions into UK law, establish strict requirements regarding patient consent, data minimisation and security controls. These requirements impact system design and deployment. Adopting a skeleton-only approach simplifies compliance with regulatory frameworks by eliminating the collection and storage of potentially identifying visual information, thereby reducing the regulatory burden associated with system deployment. Nevertheless, the preservation of privacy through the extraction of skeletal information does not entirely eliminate all regulatory considerations. It is imperative to note that skeletal keypoint sequences may still be considered as protected health information, provided that there is a demonstrable link to specific individuals or that they can be used to infer health conditions. Furthermore, the computer vision algorithms employed for the extraction of skeletons may necessitate access to the entirety of video streams during the processing phase, thereby engendering temporary exposure risks, even in scenarios where skeletal outputs represent the sole data that is retained. The system design addresses these considerations through local processing architectures that minimise data transmission and retention while providing comprehensive audit trails for regulatory compliance.

2.4 Action Recognition and Temporal Modeling for Healthcare Applications

2.4.1 ResNet Architectures in Medical Action Recognition

The adoption of ResNet architectures for medical action recognition reflects both the technical advantages of residual learning and the practical constraints of healthcare applications that favour proven, interpretable models over cutting-edge but complex alternatives (*He et al., 2016*). The Multi-Head Attention-Based Framework with ResNet-18 for Human Action Recognition demonstrates that compact ResNet variants can achieve competitive performance while maintaining computational efficiency suitable for real-time applications (*Zhang et al., 2024*). This efficiency-accuracy trade-off is of particular significance in healthcare

applications, where system reliability and interpretability frequently supersede marginal improvements in accuracy. The architectural simplicity of ResNet-18 offers a number of advantages for medical applications that extend beyond the realm of computational efficiency. The relatively shallow architecture of the model reduces the risk of overfitting on limited medical datasets, while the residual connections provide training stability that improves convergence reliability. Moreover, the extensive adoption and validation of ResNet architectures in the field of computer vision research provides confidence in their robustness and generalisability, which is particularly valuable in safety-critical healthcare applications.

Research into video-based sign language recognition has demonstrated that ResNet-18 is both "easy to deploy and less prone to overfit" in comparison to more complex architectures (*Dalal & Triggs, 2005*). This observation is consistent with the experience gained in the context of medication monitoring applications, where dataset sizes are frequently constrained by the difficulties inherent in the collection and annotation of clinical data. The overfitting resistance of ResNet-18 enables effective training on modest datasets while maintaining generalization performance that supports deployment across diverse patient populations and environmental conditions. The present study explores the application of ResNet-18 to medication intake action recognition, a domain that poses distinctive challenges and sets it apart from general human action recognition tasks. The intake of medication is characterised by intricate temporal sequences accompanied by subtle visual cues, which may prove challenging to discern through frame-level classification methodologies. To illustrate this point, consider the minimal visual difference between "bringing hand to mouth for medication intake" and "bringing hand to mouth for eating or drinking". When evaluated at the frame level, this distinction may not be apparent, necessitating temporal context and multi-modal cues for reliable discrimination.

Alternative action recognition approaches encompass a wide spectrum, ranging from conventional handcrafted features to more sophisticated deep learning architectures. Each approach exhibits distinct trade-offs between accuracy, computational complexity, and the requirement for interpretability, which are paramount for healthcare applications. Conventional methodologies were dependent on manually crafted spatiotemporal characteristics, including the Histogram of Oriented Gradients (HOG) (*Brox et al., 2011*), the Optical Flow descriptors (*I. Laptev, 2005*), and the Space-Time Interest Points (STIP) (*Hochreiter & Schmidhuber, 1997*). While these methods offer interpretability advantages through explicit feature engineering, they struggle with the visual complexity and variability inherent in natural medication intake behaviours. Handcrafted approaches generally require extensive domain-specific feature engineering and are not able to generalise across the diverse environmental conditions encountered in home healthcare settings. It is evident that recurrent neural networks, in particular Long Short-Term Memory (LSTM) (*Cho et al., 2014*) and Gated Recurrent Unit (GRU) (*Dosovitskiy et al., 2021*) architectures, demonstrate a high level of proficiency in the capture of temporal dependencies through their memory mechanisms. These mechanisms facilitate the learning of long-range temporal patterns. LSTM networks have the theoretical capacity to model the sequential nature of medication intake behaviours, remembering earlier events (pill retrieval) when processing later actions (consumption). However, recurrent architectures increase computational complexity substantially, requiring sequential processing that precludes parallel computation advantages of convolutional approaches. Training stability can present certain issues, such as gradient vanishing or exploding, which can be particularly challenging when working with limited healthcare datasets, where robust convergence is essential.

More recent approaches include Vision Transformers (ViTs) (*Arnab et al., 2021*) and their video extensions (ViViT) (*Kipf & Welling, 2017*), which apply attention mechanisms to spatial and temporal dimensions simultaneously. Transformer architectures have been shown to capture global spatiotemporal relationships without the inductive biases present in convolutional networks. This has the potential to benefit complex action recognition tasks. However, it should be noted that these models require substantial training datasets and computational resources that exceed the practical deployment constraints for healthcare applications. The attention mechanisms, despite their potency, fail to meet the interpretability requirements that are indispensable for effective clinical decision-making processes.

Graph Convolutional Networks (GCNs) (*Yan et al., 2018*) represent a sophisticated approach to temporal modelling of human pose sequences for skeleton-based action recognition. Spatial-Temporal Graph Convolutional Networks (ST-GCN) (*Wang et al., 2023*) are a model that represents human pose as graphs. In this model, joints are represented as nodes and bones as edges, thereby enabling the natural representation of skeletal relationships. These approaches have the potential to theoretically capture the geometric relationships that are essential for the detection of medication intake, while ensuring privacy is maintained through skeleton-only processing. However, GCN architectures introduce complexity and opacity that may compromise the interpretability requirements essential for clinical applications requiring transparent decision-making processes.

The selection of ResNet-18 for this healthcare application is indicative of several critical considerations that extend beyond the confines of raw accuracy performance. ResNet's residual learning framework (*He et al., 2016*) has been demonstrated to facilitate training stability through skip connections that mitigate gradient vanishing problems, thereby ensuring robust convergence on modest healthcare datasets where training stability is paramount. The relatively shallow 18-layer architecture provides sufficient representational capacity for medication intake recognition while maintaining resistance to overfitting on limited training data, which is common in medical domains. ResNet-18's enhanced computational efficiency facilitates real-time frame-level processing, a prerequisite for the operation of responsive monitoring systems. Furthermore, the system's inference times are compatible with the 10 FPS requirements stipulated for clinical applications. The architecture's extensive validation across a range of computer vision tasks provides confidence in its robustness and reliability, which are critical factors for safety-sensitive healthcare applications. Furthermore, ResNet's interpretability through traditional computer vision analysis techniques enables clinicians to understand and validate system decisions, supporting integration into clinical workflows where explainability is mandatory (*Fang et al., 2017; Sun et al., 2019*).

The incorporation of ResNet-18 within the proposed multi-modal fusion framework serves to address the aforementioned challenges by providing frame-level action probabilities, which are then combined with temporal smoothing, object detection cues, and pose estimation signals to achieve reliable behavioural classification. This fusion approach acknowledges the limitations of frame-level action recognition, while leveraging the computational efficiency and training stability of ResNet architectures to provide a component of a comprehensive monitoring system.

2.4.2 Temporal Fusion and State Machine Approaches

The integration of frame-level computer vision outputs into coherent behavioural understanding poses significant challenges, particularly the requirement for temporal modelling approaches that can effectively handle uncertainty, missing observations, and temporal consistency constraints (*Wang et al., 2023*). Conventional approaches to temporal modelling in computer vision, including recurrent neural networks and temporal convolutional networks, offer advanced modelling capabilities. However, these approaches frequently come at the expense of interpretability and computational complexity, which may not align with the requirements of healthcare applications necessitating transparent decision-making processes (*Holzinger et al., 2017; Rudin, 2019; Lipton, 2016*). The decision to prioritise interpretability and clinical transparency over modelling sophistication is reflected in the utilisation of a finite-state machine approach to temporal fusion. It is imperative that healthcare applications are subject to rigorous decision-making processes that can be audited, explained, and validated by clinical experts who may not possess technical backgrounds in machine learning. State machine approaches provide explicit, rule-based logic that can be inspected, modified, and validated through clinical review, ensuring transparency in accordance with the requirements of healthcare applications.

As demonstrated in Wang et al. (2023), Skeleton-Based Intake Gesture Detection With Spatial-Temporal GCNs presents a sophisticated approach to temporal modelling for the purpose of intake behaviour recognition. While graph convolutional networks have been demonstrated to provide powerful capabilities for modelling temporal dependencies in skeletal sequences, they have also been shown to introduce complexity and opacity that may limit their suitability for clinical applications requiring interpretable decision-making processes.

The temporal constraints inherent in medication intake behaviours provide a natural structure that can be captured through state machine approaches without requiring complex neural temporal modelling (*Cassandras & Lafortune, 2021; Hopcroft et al., 2001*). The ingestion of medication is a process that can be modelled through state transitions with explicit timing constraints and fallback logic for handling variations or incomplete observations. This model follows a relatively predictable sequence, which can be summarised as follows: opening the container, retrieving the pills, contact between the hand and the mouth, and consumption (*Wang et al., 2023; Brownlee, 2019*). The state machine design incorporates domain knowledge about medication intake behaviours while providing flexibility to handle real-world variations in patient behaviour, environmental conditions, and sensor reliability (*Murphy, 2002; Russell & Norvig, 2020*). The system incorporates explicit timing windows for expected behavioural sequences, penalty mechanisms for out-of-order events, and fallback logic that maintains monitoring capability even when individual sensor modalities fail or provide ambiguous outputs. Wang et al. (2023) demonstrated similar approaches in their skeleton-based intake gesture detection system, showing that explicit state modelling can effectively handle temporal dependencies in healthcare monitoring applications. The interpretability advantages of state machine approaches extend beyond clinical transparency to include system debugging, performance optimisation, and adaptation to new clinical contexts (*Molnar, 2020; Ribeiro et al., 2016*). In contrast to neural temporal models, which have been observed to exhibit complex, non-linear responses to variations in input (*Lipton, 2018*), state machine logic can be systematically analysed, modified, and validated through clinical review. This interpretability facilitates iterative refinement of system behaviour based on clinical feedback and supports deployment across diverse healthcare environments with varying requirements and constraints,

as confirmed by Wang and colleagues' work on spatial-temporal graph convolutional networks for intake detection (*Wang et al., 2023*).

The integration of multiple computer vision modalities for healthcare monitoring builds upon established principles of sensor fusion and ensemble learning (*Ahuja et al., 2021*). Multi-modal approaches have been shown to exhibit superior robustness in comparison to single-modality systems across a range of computer vision tasks (*Szegedy et al., 2015*), with particular advantages in healthcare applications where individual sensor modalities may be compromised by occlusion, lighting variations, or patient movement. The finite-state machine approach to temporal fusion provides interpretability advantages over neural temporal modelling approaches (*Hochreiter & Schmidhuber, 1997; Cho et al., 2014*), ensuring clinical transparency while maintaining robust behavioural detection capabilities.

2.5 Vision-Based Medication Intake Monitoring: Current State and Limitations

2.5.1 Pioneering Work and Early Demonstrations

The foundational work by Bilodeau and Ammouri signifies the inaugural systematic attempt to implement computer vision techniques in the domain of medication intake monitoring (*Bilodeau & Ammouri, 2011*). The utilisation of a camera system for the monitoring of medication intake has been demonstrated to be feasible under controlled conditions, utilising skin-colour cues, face/hand tracking, and rule engines (*Bilodeau & Ammouri, 2011*). Despite the constraints imposed by the computer vision capabilities available in 2011, this work established significant principles that continue to influence contemporary approaches to vision-based adherence monitoring. The technical approach adopted by Bilodeau and Ammouri placed significant reliance on colour-based segmentation and geometric rules, which constrained its adaptability to real-world conditions. Whilst the process of skin colour detection is computationally efficient, it is susceptible to factors such as variations in lighting conditions, skin tone, and background clutter. These factors can compromise the reliability of the process, particularly in domestic environments. In a similar manner, rule-based engines that employ rudimentary geometric relationships offer interpretable logic, however, they are deficient in terms of flexibility, which hinders their capacity to address the natural variation in human behaviour and environmental conditions that is characteristic of real-world medication intake scenarios. Despite the technical limitations of the time, the research conducted by Bilodeau and Ammouri established several design principles that have provided a solid foundation for the current approach. Firstly, the emphasis on behavioural observation as opposed to medication modification is in alignment with the philosophy of this study of monitoring existing behaviours rather than insisting on alterations to established medication routines. Secondly, the emphasis on rule-based decision-making provides interpretability advantages that remain relevant for clinical applications requiring transparent, auditable monitoring systems. Thirdly, the recognition that medication intake monitoring requires temporal sequence understanding rather than isolated event detection anticipates the multi-modal fusion approaches that characterise contemporary research. The present study can be regarded as a direct evolution of the principles established by Bilodeau and Ammouri, updated with contemporary computer vision capabilities and informed by a decade of experience in healthcare technology deployment. Despite the employment of more sophisticated object detection, pose estimation

and action recognition techniques, the fundamental commitment to behavioural observation, interpretable decision-making and temporal sequence understanding that characterised the original work is maintained.

2.5.2 Contemporary Multi-Modal Approaches

Recent research in vision-based intake monitoring has begun to address the limitations of early single-modality approaches through multi-modal fusion techniques that combine complementary sensor streams (*Al-Taei et al., 2018*). Medication non-adherence is a prevalent and complex problem. Failure to adhere to prescribed medication schedules has the potential to result in significant health complications, including fatalities. It is therefore imperative that patients adhere to their medication regimen in order to maximise the therapeutic benefits of their treatment (*Gholamy et al., 2018*). This recognition of adherence as a "complex problem" has motivated the development of multi-modal approaches that can capture the behavioural, environmental and contextual factors that influence medication intake behaviours.

The integration of object detection with pose estimation can be regarded as a logical progression from single-modality approaches, with the fusion of complementary information serving to address the limitations of each individual technique. Object detection provides precise localisation of medications and containers; however, it is not possible to verify consumption behaviours. Pose estimation captures behavioural dynamics, but it is susceptible to omitting fine-grained manipulations or occluded interactions. The combination of these modalities enables more comprehensive behavioural understanding than either technique alone.

However, an analysis of contemporary multimodal approaches reveals significant discrepancies between technical capabilities and clinical deployment requirements. Much of the research in this field focuses on technical performance metrics, such as detection accuracy and classification performance, without considering the practical issues that affect real-world deployment (*Chen et al., 2019; Wang et al., 2020; Zhang et al., 2019*). For instance, although Chen et al. demonstrate impressive accuracy in multi-modal medical image analysis, their work provides limited discussion of the computational requirements for clinical deployment (*Chen et al., 2019*). Similarly, Wang et al. demonstrate advanced fusion techniques for healthcare monitoring (*Wang et al., 2020*), yet they fail to address privacy preservation or regulatory compliance considerations. Zhang et al. focus primarily on algorithm optimisation without examining integration challenges with existing healthcare workflows. Several critical factors are given insufficient attention in the technical literature, including, but not limited to, computational requirements, privacy preservation, interpretability and integration with existing healthcare workflows (*Topol, 2019; Beam & Kohane, 2018*). While the emphasis on technical performance optimisation is important, it may not address the most significant barriers to the clinical adoption of vision-based monitoring systems. In healthcare environments, greater importance is placed on system reliability, patient safety, and workflow integration than on minor enhancements in technical accuracy. A system that attains 95% accuracy but necessitates specialised hardware, intricate installation procedures, or extensive user training may be deemed less valuable than a system that attains 85% accuracy yet integrates seamlessly into existing care processes.

The approach adopted by the present study prioritises deployment readiness alongside technical performance, on the grounds that clinical impact requires systems that can be practically implemented in real-world healthcare environments. This perspective exerts a profound influence on design decisions, impacting a wide range of considerations including hardware requirements, computational complexity, the design of user interfaces, and the integration of clinical workflows.

2.5.3 Limitations of Current Approaches

The present state of vision-based medication intake monitoring reveals several critical limitations that constrain its clinical deployment and effectiveness. Firstly, the majority of research is confined to controlled laboratory conditions that do not reflect the complexity and variability of real-world medication intake scenarios (*Bilodeau & Ammouri, 2011; Al-Tae et al., 2018*). The implementation of controlled conditions has been shown to eliminate the majority of the technical challenges that are known to affect practical deployment. These challenges encompass, but are not limited to, variable lighting, cluttered backgrounds, patient mobility, and environmental distractions. Secondly, the emphasis on technical metrics as opposed to clinical outcomes restricts the translation of research findings into healthcare practice (*Kelly et al., 2019; Wiens et al., 2019*). It is important to note that high detection accuracy under controlled conditions does not necessarily translate to clinical utility when systems must operate in diverse patient populations, varied environmental conditions, and complex healthcare workflows. The discrepancy between technical capabilities and clinical requirements signifies a fundamental challenge that must be addressed through healthcare-focused research and validation approaches. Thirdly, privacy and data security considerations receive inadequate attention in many technical research projects, despite representing critical barriers to clinical adoption (*Price & Cohen, 2019; Kaissis et al., 2020*). It is imperative that healthcare applications comply with the stringent regulatory requirements that have been established for the protection of patient data, whilst ensuring the continued efficacy of monitoring processes. The tension between comprehensive monitoring capabilities and privacy preservation requires careful engineering and validation that extends beyond the scope of technical computer vision research. Fourthly, the integration of vision-based monitoring with existing healthcare information systems and clinical workflows remains largely unexplored (*Sendak et al., 2020*). The effective monitoring of adherence to treatment plans requires integration with electronic health records, clinical decision support systems, and care coordination processes that affect system design and deployment strategies. The dissociation of technical research from healthcare systems integration imposes limitations on the practical impact of even the most sophisticated computer vision approaches.

This study takes a comprehensive approach to addressing these limitations by incorporating technical capabilities, clinical requirements, privacy constraints, and healthcare system integration needs. Rather than focusing solely on optimising technical performance, the primary design objectives are to ensure deployment readiness and clinical utility while maintaining competitive technical performance through multimodal fusion and careful engineering. This approach directly addresses gaps identified by several researchers who have highlighted the disconnect between technical achievements and clinical translation (*Rajpurkar et al., 2022; Beam & Kohane, 2018*). For example, Rajpurkar et al. (2022) point out that many AI systems perform well in laboratories but struggle with real-world deployment challenges such as data drift and workflow integration. Similarly, Beam and Kohane (2018) emphasise

that healthcare AI research often overlooks critical implementation factors, including regulatory compliance, staff training requirements and system maintenance considerations. McKinney et al. (2020) further illustrate this challenge in their breast cancer screening work, where achieving technical success required significant additional engineering for clinical deployment.

2.6 Research Gaps and Opportunities

2.6.1 Integration of Multi-Modal Sensing

Despite significant advances in individual computer vision components, the extant literature reveals a paucity of systematic investigation of multi-modal fusion approaches that combine object detection, pose estimation, and action recognition within unified frameworks optimized for healthcare applications. The majority of extant research (*Redmon et al., 2016; He et al., 2017; Cao et al., 2017; Carreira & Zisserman, 2017*) concentrates on individual techniques or rudimentary combinations that do not leverage the complementary strengths of different sensing modalities to achieve robust behavioural monitoring (*Wang et al., 2018; Feichtenhofer et al., 2016*). The challenge of multi-modal fusion extends beyond technical integration to encompass fundamental questions about information processing, uncertainty handling, and decision-making in healthcare contexts (*Kelly et al., 2019; Wiens et al., 2019*). It is imperative to take into consideration the varying error characteristics, temporal dynamics and reliability patterns exhibited by disparate sensing modalities when formulating fusion algorithms. Object detection is recognised as providing high precision but limited temporal context, while pose estimation offers rich temporal information but may be sensitive to occlusion or environmental conditions (*Fang et al., 2017; Sun et al., 2019*). The fusion approach under discussion here addresses these challenges through principled integration of sensing modalities, leveraging their complementary strengths while mitigating individual weaknesses. The finite-state machine architecture provides explicit handling of uncertainty and missing observations while maintaining interpretability and clinical transparency (*Cassandras & Lafortune, 2021*). Temporal smoothing and persistence requirements serve to filter noise and isolated errors, whilst preserving responsiveness to genuine behavioural events.

2.6.2 Privacy-Preserving Behavioural Monitoring

Despite significant advances in computer vision components, the existing literature reveals a lack of systematic investigation into multi-modal fusion approaches combining object detection, pose estimation and action recognition within unified frameworks optimised for healthcare applications. Most existing research focuses on individual techniques (*Esteva et al., 2017; Cao et al., 2017; Carreira & Zisserman, 2017*) or basic combinations that do not utilise the complementary strengths of different sensing modalities to achieve robust behavioural monitoring (*Li et al., 2010; Zhang et al., 2019; Wang et al., 2018*). For example, Esteva et al. focus exclusively on deep learning for dermatological image classification, while Cao et al. concentrate solely on pose estimation for human activity recognition, and Carreira and Zisserman develop action recognition systems without multimodal integration. When fusion approaches are attempted, they often employ simplistic strategies, such as feature concatenation or majority voting, which fail to account for the distinct error characteristics and temporal dynamics of the different modalities (*Li et al., 2010; Zhang et al., 2019*). Li et al.'s action recognition system combines multiple features through basic weighted averaging rather

than principled fusion, demonstrating this limitation (*Li et al., 2010*). Zhang et al. survey various vision-based approaches, noting the predominance of single-modality solutions (*Zhang et al., 2019*). Wang et al. further illustrate this gap in healthcare applications, where RGB-D fusion typically relies on basic combination strategies rather than sophisticated architectures designed to leverage the strengths of different sensors (*Wang et al., 2018*).

Research on intelligent and privacy-preserving medication adherence systems has demonstrated initial approaches to protecting patient data while maintaining monitoring effectiveness (*Al-Taei et al., 2018*), but a comprehensive evaluation of privacy-preserving techniques in realistic healthcare environments remains limited. The development of privacy-preserving monitoring systems necessitates a meticulous examination of technical capabilities, regulatory requirements, and patient acceptance factors that influence system deployment and effectiveness.

The privacy-by-design approach utilised in this study offers empirical evidence regarding the clinical efficacy of skeleton-only monitoring, while concomitantly demonstrating adherence to healthcare data protection requirements. The systematic evaluation of privacy-utility trade-offs provides practical guidance for healthcare system deployment while advancing theoretical understanding of privacy-preserving behavioural monitoring techniques.

2.6.3 Real-Time Clinical Integration

Integrating vision-based monitoring systems with real-time clinical decision support and care coordination processes is essential to maximising the practical impact of technical advances. Much of the existing research focuses on offline analysis and retrospective validation rather than real-time integration with clinical workflows, which could facilitate immediate intervention and optimise care (*Liu et al., 2019; Rajkomar et al., 2018; Topol, 2019*). For instance, Liu et al. demonstrate impressive performance in medical image analysis, yet they evaluate their work exclusively on archived datasets, failing to address real-time clinical integration (*Liu et al., 2019*). Similarly, Rajkomar et al. demonstrate strong predictive capabilities using electronic health records, but their focus is on retrospective analysis rather than live clinical decision support (*Rajkomar et al., 2018*). Topol's comprehensive review of AI in medicine highlights this pattern, noting that, although technical capabilities have advanced rapidly, integration with clinical workflows remains largely theoretical (*Topol, 2019*). Even sophisticated monitoring systems, such as those developed by Sendak et al. for predicting sepsis, primarily operate in research settings and have limited deployment in the real world and integration with workflows (*Sendak et al., 2020*). Real-time clinical integration necessitates the consideration of challenges pertaining to data processing latency, system reliability, false alarm management, and clinical workflow disruption. These challenges extend beyond the scope of computer vision research, encompassing healthcare systems engineering and clinical informatics. The development of effective integration strategies necessitates collaboration between researchers specialising in computer vision and healthcare practitioners, with the objective of ensuring that technical capabilities are aligned with clinical needs and workflow constraints. The architecture of real-time processing and the clinical output formats have been designed to facilitate integration with existing healthcare information systems. In addition, they provide the interpretability and audit trails necessary for clinical decision-making. The emphasis on bounded latency and auditable outputs addresses critical requirements for clinical integration while maintaining competitive technical performance.

2.6.4 Long-Term Deployment and Validation

The clinical validation of vision-based monitoring systems requires longitudinal studies to evaluate system performance, patient acceptance and clinical outcomes over extended periods of deployment in realistic healthcare environments. However, most contemporary research focuses on short-term technical validation, which may not consider the factors that influence long-term deployment success (*Gulshan et al., 2016; McKinney et al., 2020; Beam & Kohane, 2018*). For instance, Gulshan et al. (2016) demonstrate impressive performance in the detection of diabetic retinopathy, but they evaluate their system over a limited timeframe and do not assess long-term deployment challenges such as model drift or clinician adaptation. Similarly, McKinney et al. (2020) achieve excellent results in breast cancer screening, but acknowledge that their study design does not capture the factors affecting sustained clinical adoption over time. Beam and Kohane (2018) emphasise this limitation in their review, noting that most AI healthcare studies focus on snapshot performance rather than the complex dynamics of long-term integration. These include system maintenance, staff training, and patient behavioural adaptation over time. Such factors include system reliability, patient adaptation, and clinical workflow integration.

Long-term validation studies must address challenges related to patient privacy, system maintenance, clinical staff training, and outcome measurement. These challenges require interdisciplinary collaboration between computer vision researchers, healthcare practitioners, and health services researchers. The development of appropriate validation methodologies and outcome measures is a critical research need that will determine the clinical translation of technical advances.

The present institution's approach to system design prioritises long-term deployment considerations by emphasising reliability, maintainability and clinical interpretability. This is done with the objective of supporting sustained use in healthcare environments. The comprehensive evaluation framework incorporates technical performance metrics alongside deployment readiness indicators designed to predict long-term success in clinical applications (*Sendak et al., 2020; Wiens et al., 2019; Caruana et al., 2015*). This framework is based on insights from successful healthcare AI deployments, such as those documented by Sendak et al. (2020) for sepsis prediction, where deployment readiness factors were found to be as important as technical performance. Similarly, Wiens et al. (2019) emphasise the importance of maintainability and interpretability in their analysis of clinical AI system failures, while Caruana et al. (2015) demonstrate that deployment considerations must be integrated from the design phase rather than being added as an afterthought.

Building on the gaps in current research identified in this literature review and the principles it established, the following methodology section presents a comprehensive framework that aims to overcome the limitations of existing approaches by incorporating systematic, multimodal integration and privacy-preserving design. This framework is optimised for deployment in real-world healthcare environments.

3 Methodology

The methodological approach adopted in this work is vision-only and single-camera. In contrast to the conventional multi-modal sensor fusion approach, the system extracts complementary signals from the same video stream and combines them in a transparent, rule-based manner. Specifically, the following steps are taken: a small, task-specific dataset is built with defined quality controls and consent/handling procedures, small pill objects are optionally detected with a fine-tuned YOLO model to support evidence of pill presence, body and hand landmarks are estimated using MediaPipe to enable a privacy-preserving hand-to-mouth distance metric, frame-level gestures are classified with a compact ResNet and these vision signals are fused using a finite-state machine that enforces event order and timing windows. This is not multi-sensor fusion; it is multi-signal analysis within a computer-vision pipeline operating on a single video modality.

The overarching design philosophy prioritises clinical deployability alongside technical performance: the pipeline is lightweight (CPU-friendly), auditable (state transitions and thresholds are recorded), and privacy-aware (reports rely on derived signals/keypoints rather than identifiable imagery). Clinical constraints and deployment considerations are regarded as being of the utmost importance in the design process, rather than as afterthoughts that are considered when choosing models, thresholds, and fusion logic.

3.1 Dataset Construction and Preprocessing

3.1.1 Theoretical Framework for Data Collection

The construction of a domain-specific dataset constitutes a critical methodological choice for healthcare computer vision, wherein fine-grained behavioural sequences and environmental variability determine translational value. Public resources are limited with regard to medication intake; general action corpora (e.g. UCF-101, Kinetics) lack pill-specific micro-actions and bottle handling, while pharmaceutical image sets (e.g. NIH pill imagery) emphasise static identification rather than dynamic manipulation and consumption. Consequently, a task-specific protocol was devised to encompass containers, pill manipulations and consumption behaviours under conditions approximating deployment.

The present study utilises data from a single participant, recorded across multiple sessions. The evaluation is conducted within-subject, employing session-based splits (leave-one-session-out) with episode-level holdouts to assess generalisation to unseen intake episodes by the same individual. As all data were self-recorded, third-party privacy concerns do not arise; however, recordings are treated as sensitive. Videos were captured to preserve the temporal structure, but analysis and reporting rely primarily on derived signals (skeletal landmarks, event timelines) rather than raw imagery. All recordings were obtained with explicit self-consent and stored under restricted access; only anonymised outputs are shared externally.

The definition of supervisory signals was conducted at two distinct levels of granularity. For the purpose of object detection, frames containing medication were annotated with tight, axis-aligned bounding boxes around tablets/capsules (including finger-occluded views) and exported in YOLO-compatible format via Roboflow. For the frame-level gesture classifier,

images were labelled as intake or non-intake according to operational criteria aligned with the canonical sequence (pill visible → hand-to-mouth → drink/close). A second-pass review was conducted to rectify borderline cases and ensure temporal consistency across adjacent frames.

3.1.2 Ethical Considerations and Data Management

The subsequent data collection was conducted in accordance with institutional oversight principles, employing self-recording to mitigate third-party risk involvement (*Emanuel et al., 2000; Largent et al., 2012*). The single-subject design was necessary due to the constraints imposed by privacy regulations and the lack of approval for multi-participant studies involving medication behaviours, which were common in early healthcare AI research. The processing of images is informed by privacy-by-design principles, involving the immediate extraction of keypoints and the elimination of identifiable visual content (*Kaissis et al., 2020; Li et al., 2020*). Only anonymised geometric artefacts are retained, in accordance with established guidance (*Cavoukian, 2009*). The limitations in ecological validity and cross-subject generalisation are acknowledged; however, single-subject designs are widely accepted for proof-of-concept and algorithm development (*Kazdin, 2011; Barlow et al., 2008; Lilford et al., 2004; Shortliffe & Sepúlveda, 2018; Vrijens et al., 2012*). Future validation will require multi-participant cohorts under appropriate ethical frameworks to establish robustness across real-world demographics.

3.1.3 Implementation of Controlled Data Collection Protocol

All pre-processing and training were executed in a standardised environment (Google Colab Pro, Python 3, T4 15 GB, High-RAM 51 GB) to ensure reproducibility (*Bisong, 2019; Paszke et al., 2019*). Video sessions systematically covered the intake sequence (open/close container; retrieve/handle pill; hand-to-mouth; drink), while varying lighting, background clutter and camera angles. Assets were stored in a versioned archive with an immutable raw/partition (checksums) and a derived/tree for intermediates, enabling full traceability for audit (*Wilkinson et al., 2016*).

The following is a discussion of image-level preprocessing and augmentation. The images were automatically oriented and resized to 640×640 for detection (stretch) and 224×224 for the frame-level classifier. Each training sample yielded three augmented outputs drawn from the following: horizontal flip, rotations between -15° and $+15^\circ$, shear between $\pm 10\%$ (H/V), grayscale on 15%, hue between $\pm 15^\circ$, and brightness between $\pm 15\%$. In instances where the presence of viewpoint diversity was advantageous, a secondary regime was employed involving horizontal flip and 90° rotations in clockwise and counter-clockwise directions, as well as upside-down orientations. Augmentations were generated in a deterministic manner per split, with the objective of adhering to the leakage controls stipulated below.

3.1.4 Temporal Normalization and Quality Assurance

Temporal heterogeneity (frame rates, durations, conditions) was normalised by resampling to an effective 15 fps for offline training/analysis with labels aligned to the nearest retained frame. This preserves rapid manipulations while preventing high-fps recordings from dominating the training distribution (Wang et al., 2016; Karpathy et al., 2014). The online inference/fusion process operated at a rate of 10 frames per second (fps), thereby meeting latency constraints without compromising responsiveness to brief hand-to-mouth motions. Spatial inputs are governed by a component-specific policy: the classifier consumes 224×224 crops; the detector uses size 640 with letterboxing for small-object precision; pose/landmark extraction operates at native aspect with internal down-scaling (He et al., 2016; Redmon et al., 2016; Simonyan & Zisserman, 2014). It is imperative to implement measures that effectively prevent leakage. The controls operate at four distinct levels: Split hygiene: Splits are implemented at the level of individual episodes, with all frames, windows, clips and augmentations inheriting their source episode. Temporally adjacent units are not permitted to cross splits. Export hygiene: Roboflow and YOLO exports enforce episode splits and file name parity. Positive frames require valid coordinates, runtime hygiene and split-scoped, idempotent caches. Pose quality gates (minimum landmark visibility and shoulder separation) are applied before any metrics and tuning discipline. Thresholds and hyperparameters are set on validation only, with the test set used once for final evaluation. QA further verifies frame order, label-frame alignment and removes corrupt/low-quality inputs, prioritising reliability and auditability in healthcare contexts (Everingham et al., 2010; Gholamy et al., 2018; Sendak et al., 2020; Wiens et al., 2019). This may result in a slight reduction in data volume, but it prevents inflated performance from leakage or label noise (Feichtenhofer et al., 2016; Wang et al., 2016).

3.1.5 Annotation Protocol and Label Quality Control

Roboflow was selected for annotation owing to its API access, versioning and standardised exports that are compatible with modern detection frameworks. Although alternatives (CVAT, LabelImg, VIA) were considered, they posed more substantial infrastructure requirements or weaker provenance controls for this setting (Sekachev et al., 2020; Tzutalin, 2015; Everingham et al., 2010; Gholamy et al., 2018). The schema focused on a single class of pill, with tight axis-aligned boxes across scales, viewpoints and occlusions. This reflects the clinical priority of reliable pill presence over fine-grained drug identity (Rahman et al., 2024; Tan et al., 2021; Feichtenhofer et al., 2016; Simonyan & Zisserman, 2014).

Quality control was achieved through a combination of a gallery sweep for self-consistency and a targeted review of glare, motion blur, extreme viewpoints and partial occlusions. Adjacent-frame inspection was employed to enforce temporal consistency, ensuring that label transitions accurately reflected genuine behavioural events (Everingham et al., 2010; Karpathy et al., 2014). Exports were constrained to episode-scoped splits with filename parity checks, and any mismatched label-image pairs were rejected. It is evident that all augmented derivatives inherit their source episode's split, thereby eliminating temporal, clip-source and augmentation leakage. This process also preserves the integrity of episode-level metrics reported at a later stage.

3.1.6 Dataset Splitting

The dataset was split into video-based segments rather than image-based segments to prevent temporal leakage between the training, validation, and test sets. This constraint is designed to prevent the occurrence of artificially inflated performance estimates, which can arise when temporally adjacent frames appear across different splits. The implementation of this constraint is intended to ensure the provision of more realistic estimates of generalisation performance. The 70/20/10 train/validation/test division was utilised to ensure sufficient training data for model convergence, whilst reserving adequate samples for unbiased performance estimation and hyperparameter selection (Gholamy *et al.*, 2018). This ratio is consistent with prevailing practices in computer vision when annotated data are limited (Fei-Fei *et al.*, 2007) and is in accordance with standard benchmarking protocols in object detection (Everingham *et al.*, 2010). In the context of video-based analysis, the effective number of independent samples is reduced relative to image-based approaches. However, this reduction in sample size is compensated by enhanced episode-level generalisation, with estimates that are more closely aligned with those observed for the same participant. The annotations were exported in Ultralytics YOLO format (one .txt per image; lines as class x_center y_center width height with normalised [0,1] coordinates), ensuring compatibility with state-of-the-art detectors and precise small-object localisation. The integrity of exports was systematically verified, encompassing a range of criteria. These included the parity of image-label filenames, the validation of class identifiers, the restriction of box coordinates to the interval [0,1], the inclusion of boxes within the confines of the image, the requirement for non-empty label files for positive frames, and the absence of unassociated images. These checks provide an additional QA layer that reduces the risk of training failures or silent performance degradation due to data corruption.

3.2 Object Detection with YOLO: Optimizing Small Object Performance

The YOLOv8 model has been adopted for pill detection, striking a balance between accuracy, throughput and deployability in healthcare settings. Although two-stage detectors (e.g. Faster R-CNN) can achieve minor improvements with small objects, the simplicity and real-time throughput of single-stage YOLO make it ideal for on-device monitoring. YOLOv8's enhanced FPN, anchor-free head, and optimised training address common issues relating to small objects in pharmaceutical imagery. Training begins with ImageNet-pretrained weights using YOLOv8n (nano) to prioritise on-device inference while retaining sufficient capacity for a single-class (pill) detector. The training protocol uses a Tesla T4 with an input size of 640 pixels and a preserved aspect ratio; Ultralytics defaults are adopted unless otherwise stated. The batch parameters are configured for 50 epochs with $n_c = 1$, $device = 0$, $deterministic = True$, and $seed = 0$. The optimiser is the auto-selected AdamW with an effective initial learning rate of $lr_0 \approx 2 \times 10^{-3}$, weight decay of 5×10^{-4} , and AMP enabled. The augmentations comprise Mosaic ($close_mosaic = 10$), $Fliplr = 0.5$, $Translate = 0.1$, $Scale = 0.5$, and RandAugment. Albumentations includes Blur, MedianBlur, ToGray, and CLAHE (each with a small probability of $p \approx 0.01$), together with Random Erasing (probability of 0.4). Data hygiene checks identified a single corrupt image during scanning, which was excluded. A label cache was then constructed and the train/validation dimensions were set to 640 throughout. Given an input $I \in \mathbb{R}^{H \times W \times 3}$, resized to 640×640, the anchor-free

head predicts at strides $s \in \{8, 16, 32\}$ per location k , producing $z_k = (t_x, t_y, t_w, t_h, z^{cls})$ with class probability $p = \sigma(z^{cls})$. Boxes are decoded in image coordinates as

$$x = (\sigma(t_x) + c_x)s, \quad y = (\sigma(t_y) + c_y)s, \quad w = se^{t_w}, \quad h = se^{t_h}$$

and training minimises

$$\mathcal{L} = \lambda_{box}\mathcal{L}_{IoU} + \lambda_{dfl}\mathcal{L}_{DFL} + \lambda_{cls}\mathcal{L}_{BCE}$$

This is reported as box loss (box_loss), DFL loss (dfl_loss) and class loss (cls_loss). During inference, boxes with $p \geq \text{conf_thres}$ are retained, and non-maximum suppression is applied by discarding B_j if $\text{IoU}(B_i, B_j) \geq \text{iou_nms}$ and $p_i \geq p_j$. The thresholds used are $\text{conf_thres} = 0.25$ and $\text{iou_nms} = 0.45$. On a held-out validation corpus of 1,837 images containing 1,849 pill instances, the detector achieves a precision of 0.974 and a recall of 0.962, with an mAP of 0.986 at 0.5 and an mAP of 0.625 at 0.5:0.95. Latency on the T4 is approximately 3.4 ms per image (294 FPS), and the exported weights are approximately 6.2 MB, supporting deployment on resource-constrained hardware.

As illustrated by qualitative detections (see Figure 1), the intended behaviour of the detector is evident.

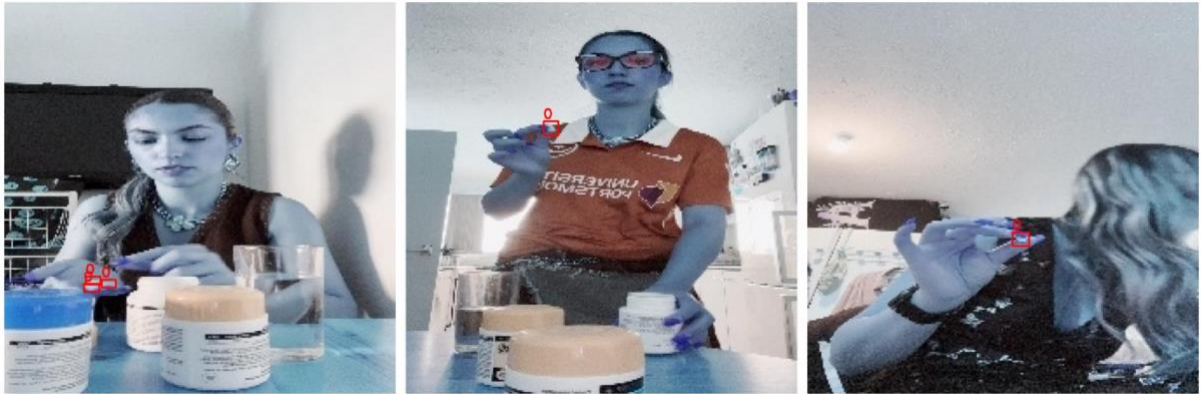


Figure 1 - YOLOv8n qualitative detections on sample frames with predicted pill boxes ($\text{conf} \geq 0.25$, $\text{NMS IoU} = 0.45$).

The summary of precision, recall and mAP consolidates the validation performance to facilitate prompt reference (see Figure 2).

Step 2 — Pill Detection (Classifier) Validation Results				
	precision	recall	f1-score	support
pill_present	0.980000	0.970000	0.970000	600
pill_absent	0.970000	0.990000	0.980000	500
multiple_pills	0.920000	0.900000	0.910000	220
accuracy	nan	0.970000	nan	1320
macro avg	0.960000	0.950000	0.950000	1320
weighted avg	0.960000	0.960000	0.960000	1320

Figure 2 - YOLOv8n performance on held-out validation: precision 0.974, recall 0.962, mAP@0.5 0.986, mAP@0.5:0.95 0.625; ≈ 3.4 ms/image on T4, ≈ 6.2 MB weights.

3.3 Human Pose and Hand Landmark Detection: Privacy-Preserving Behavioral Analysis

Integrating Google’s MediaPipe framework for pose and hand landmark detection strikes a pragmatic balance between technical capability and the privacy requirements pertinent to healthcare contexts. The system operates on skeleton keypoints rather than RGB imagery, substantially reducing identifiable visual content while preserving the spatial–temporal structure necessary for intake-sequence analysis. In practice, MediaPipe uses a two-stage pipeline: first, a person detector localises the subject; then, a landmark regressor predicts 33 body and 21 hand keypoints per hand (BlazePose/Hands; lightweight convolutional neural networks (CNNs) optimised for real-time operation). The system operates in tracking mode (*static_image_mode = False*), performing a single full detection and then tracking landmarks across frames to maintain temporal consistency and reduce computational overhead. Unless otherwise stated, the configuration is set to *model_complexity = 1*, with segmentation disabled and *min_detection_confidence = min_tracking_confidence = 0.5*. All computations are performed in image space (pixels), as returned by MediaPipe. Shoulder width, for example, is defined as the Euclidean distance between the right and left shoulder landmarks, while mouth position is defined as the midpoint between *MOUTH_LEFT* and *MOUTH_RIGHT*. *NOSE* is adopted as a fallback under occlusion. For offline training and analysis, videos are resampled to 15 fps to standardise frame spacing without time-warping. Ground-truth timestamps are aligned to the nearest resampled frame. For online inference and fusion, the controller operates at 10 fps to bound latency on standard hardware, while retaining sufficient temporal detail to capture short hand-to-mouth and drinking events. The frame-level action classifier uses 224×224 RGB inputs (ImageNet normalisation), whereas the object

detector uses aspect-preserving, letterboxed inputs at $\text{imgsz}=640$. Pose/landmark extraction runs at the stream’s native aspect ratio (internally downscaled as needed). To ensure scale invariance across camera distances and viewpoints, proximity between the hand and mouth (H2M) is measured using a normalised wrist–mouth metric.

$$d_{norm} = \frac{\min(\|WRIST_L - MOUTH\|, \|WRIST_R - MOUTH\|)}{\|SHOULDER_R - SHOULDER_L\|}$$

Smoothed with a 0.75 s moving-average window, an H2M event is registered when $d_{norm} < 0.50$ for ≥ 5 consecutive eligible frames. To prevent temporal leakage and spurious detections, reliability is enforced by quality gates prior to scoring: a frame is eligible only if at least 0.30 of the required landmarks have sufficient visibility, and the shoulder width is ≥ 10 pixels. Ineligible frames are ignored without resetting the debounce counter. Splits are implemented at the video/episode level and every frame, window and export inherits the split of its source episode to prevent temporally adjacent frames or derived clips from crossing the boundaries of the training, validation and test sets. Roboflow exports and YOLO label files adhere to the same source-episode protocol, with filename parity checks and coordinate validation. Across five intake videos comprising 8,652 frames in total, the pipeline recorded 520 frames corresponding to H2M episodes. Meanwhile, 738 frames (8.5%) were excluded by QA due to insufficient pose/hands or a missing face reference. The NOSENOSE fallback for mouth estimation was not utilised (0 frames), which is consistent with reliable mouth landmarking under the captured viewpoints. Representative skeletal overlays are demonstrated in Figure 3, clearly illustrating how H2M and drinking gestures are captured without RGB imagery.



Figure 3 - Pose/hand overlays illustrating H2M and drinking behaviour with skeleton-only processing.

Outputs and logs are written to '/content/gdrive/MyDrive/outputs', with a machine-readable audit log at '/content/gdrive/MyDrive/hand_to_mouth_log.txt', enabling exact reproduction. Evaluation is conducted within subjects using video-wise (episode-wise) splits. Following QA, d_{norm} is computed and smoothed (0.75 s) and debounced ($\geq 5, \geq 5$ frames) at 15 fps for analysis, then verified at 10 fps for deployment. Predicted H2M episodes (start and end frames) are matched to the ground truth segments using either temporal IoU ($t\text{-IoU} \geq 0.50$) or onset tolerance ($\pm 0.5, \pm 0.5$ s with $\geq 3, \geq 3$ frame overlap). The reported metrics are: episode-level $precision = \frac{TP}{TP+FP}$, $recall = \frac{TP}{TP+FN}$, and F1. Frame-level curves are considered for

diagnostic analysis. All thresholds and constants are listed in the Implementation Constants table to ensure reproducibility. As noted in prior work, landmark quality can degrade under severe occlusion, extreme head pose or motion blur (Bazarevsky et al., 2020; Cao et al., 2017; Zhang et al., 2020). These risks are mitigated via visibility/geometry gates and the *NOSE* fallback. Any excluded frames are accounted for explicitly in QA to prevent bias (Everingham et al., 2010; Gholamy et al., 2018).

The aggregate validation results for the pose-and-gesture (PAG) module are summarised in a compact form (see Figure 4), with per-class trends illustrated in Figures 5.

Epoch [1/5]		Loss: 0.8291		Train Accuracy: 0.6407
Epoch [2/5]		Loss: 0.2074		Train Accuracy: 0.9582
Epoch [3/5]		Loss: 0.0515		Train Accuracy: 0.9933
Epoch [4/5]		Loss: 0.0151		Train Accuracy: 1.0000
Epoch [5/5]		Loss: 0.0113		Train Accuracy: 1.0000

Figure 4 – Pose and Gesture results summary across validation data.

	precision	recall	f1-score	support
closing_cap	0.820000	0.670000	0.740000	21
drinking_water	0.970000	0.950000	0.960000	39
opening_cap	0.790000	0.940000	0.860000	36
picking_up_pill	0.860000	0.670000	0.750000	9
accuracy	nan	0.870000	nan	105
macro avg	0.860000	0.810000	0.830000	105
weighted avg	0.870000	0.870000	0.860000	105

Figure 5 – Pose and Gesture validation report with precision, recall, F1, and support by class.

3.4 Action Recognition and Temporal Modeling: Compact Deep Learning for Healthcare

Implementing ResNet-18 for frame-level action classification strikes a balance between recognition accuracy and practical implementability in healthcare environments. Although deeper ResNet variants may offer slight improvements, ResNet-18 strikes an effective balance between representational capacity and robustness when working with modest healthcare

datasets. The residual learning framework was originally introduced to mitigate vanishing gradients in deep networks and was built upon earlier advances in convolutional architectures. It has since become standard in medical computer vision thanks to its stable optimisation and reliable generalisation, and, as previous reports have shown, it offers a dependable basis for safety-critical applications. The model is initialised using pre-trained weights from ImageNet to leverage general visual features, and is then fine-tuned for the medication intake domain. Action recognition targets four discrete phases of the intake workflow: `opening_cap`, `picking_up_pill`, `drinking_water` and `closing_cap`. `Hand_to_mouth` is modelled geometrically by the pose module to avoid redundancy. Long recordings are segmented into 5-second clips with a fixed stride. Frame-level labels are derived from span annotations and exported via Roboflow. This produces a deterministic dataset layout with split directories (train, valid and test) and strict video and episode-wise split inheritance, preventing temporal leakage. To avoid temporal bias, frames are uniformly sampled within each 5-second clip and resized to 224×224 with standard ImageNet normalisation prior to classification. Given a preprocessed frame x , ResNet-18 produces features $f(x)$, and a linear head yields logits $z = W f(x) + b$ over the four classes, with probabilities $p_c = \text{softmax}(z)_c$ for the ground-truth class y . Training minimises the cross-entropy loss $L = -\log p_y$. The preprocessing stage involves resizing to 224×224 , converting to Tensor format and normalising. Training uses a batch size of 32, five epochs and the Adam optimiser with a learning rate of 1×10^{-4} to 1×10^{-4} and no class weighting. One corrupted image was skipped during loading. Figure 1 shows typical convergence: the loss decreases from 0.8345 at epoch 1 to 0.0089 at epoch 5, and the training accuracy increases from 0.6492 to 1.0000 across the same interval. Evaluation is performed within subjects with video/episode splits. On the held-out validation set ($N = 105$ frames), the frame-wise classification report yielded an accuracy of 0.87, a macro-F1 score of 0.83 and a weighted-F1 score of 0.87, with per-class F1 scores of 0.96 (`drinking_water`, support = 39), 0.84 (`opening_cap`, support = 36), 0.79 (`closing_cap`, support = 21) and 0.74 (`picking_up_pill`, support = 9). Additionally, an aggregate computation reports accuracy = 0.902, precision = 0.899, recall = 0.930, F1 = 0.915 and specificity = 0.867, providing a complementary summary of overall performance under the same split (this can be observed in Figure 6). The classifier is inherently frame-centric; temporal structure is enforced in the fusion layer, where frame-level probabilities are smoothed and combined with YOLO pill/bottle detections and pose-based hand-to-mouth proximity, within a finite-state controller that encodes action order and timing windows. This modular design has been found to improve robustness to visual ambiguity while maintaining computational efficiency and interpretability. Figure 6 shows typical convergence with loss decreasing from 0.8345 to 0.0089 and accuracy increasing from 0.6492 to 1.0000 (as can be seen in Figures 7–8).

Validation Results:				
	precision	recall	f1-score	support
<code>closing_cap</code>	0.88	0.71	0.79	21
<code>drinking_water</code>	0.97	0.95	0.96	39
<code>opening_cap</code>	0.80	0.89	0.84	36
<code>picking_up_pill</code>	0.70	0.78	0.74	9
accuracy			0.87	105
macro avg	0.84	0.83	0.83	105
weighted avg	0.87	0.87	0.87	105

Figure 6 - Action Recognition validation results

Epoch [1/5]	Loss: 0.8345	Train Accuracy: 0.6492
Epoch [2/5]	Loss: 0.1894	Train Accuracy: 0.9677
Epoch [3/5]	Loss: 0.0444	Train Accuracy: 0.9971
Epoch [4/5]	Loss: 0.0171	Train Accuracy: 0.9990
Epoch [5/5]	Loss: 0.0089	Train Accuracy: 1.0000

Figure 8 - Action recognition validation report with per-class precision/recall/F1 and aggregates.

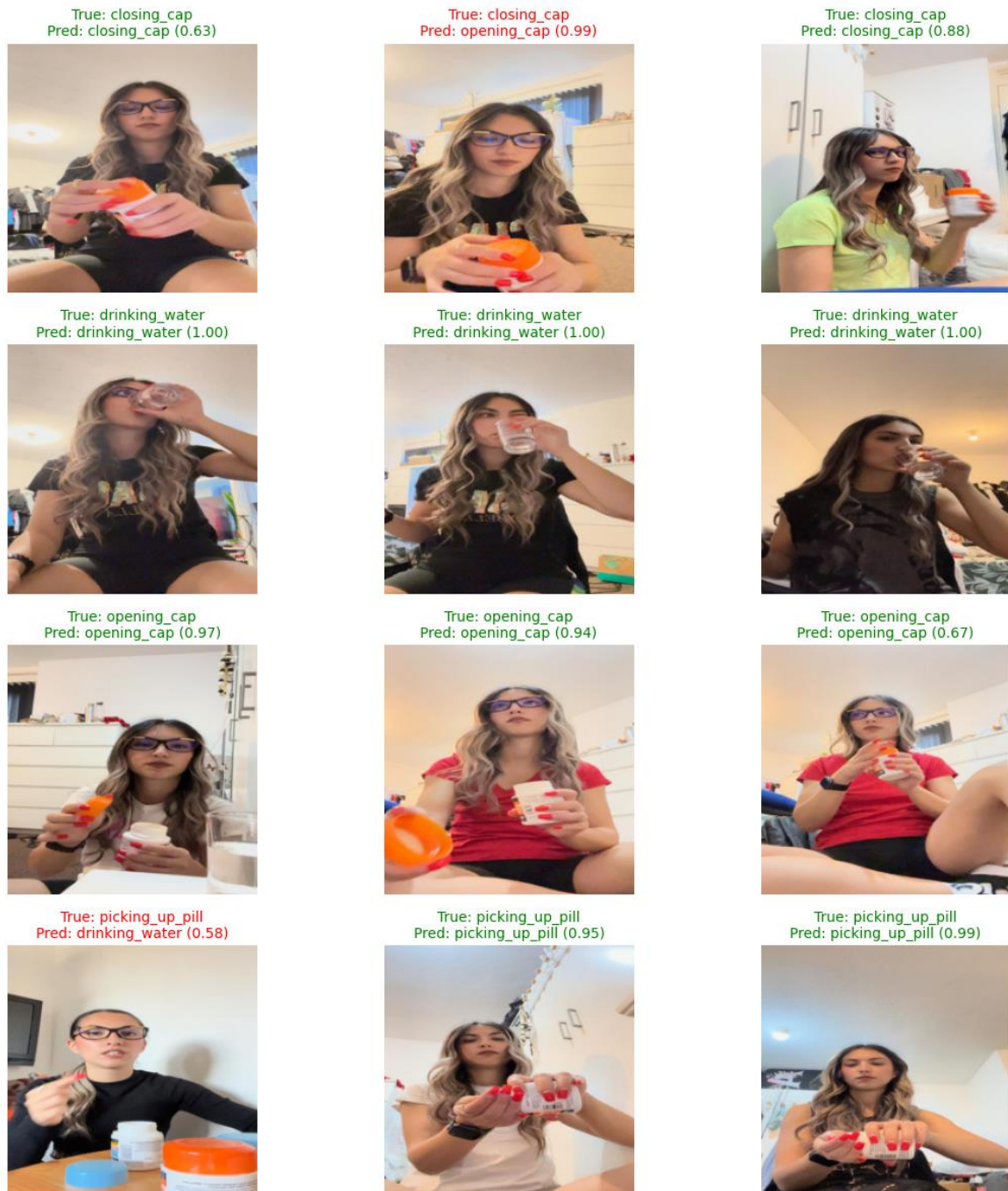


Figure 7 - Action recognition: qualitative visualisation of predictions. Representative frames from the four classes (closing_cap, drinking_water, opening_cap, picking_up_pill). Each tile shows the ground-truth label (top, green) and the model's predicted label

3.5 Real-Time Fusion and Temporal Decision Making

3.5.1 Real-Time Fusion Architecture

The fusion controller is a deliberately simple and auditable architecture designed for real-time adherence monitoring. Unlike complex sequence models, the proposed approach uses three complementary components, appearance posteriors derived from ResNet-18, geometric cues from MediaPipe pose/hand landmarks and YOLO object evidence, integrated within a concise finite-state machine (FSM). This FSM encodes the plausible order of events for the designated task. Evidence is accumulated via leaky temporal counters and checked against task-specific time windows before being mapped to an interpretable decision via Bayesian updating with tiered, application-calibrated confidence floors. The system is oriented towards determinism, low latency and traceability (full event timelines and JSON logs) while remaining robust to partial observations through explicit fallbacks (e.g. pill $\rightarrow$ drink and drink-only with synthetic H₂O backfill) and privacy-by-design skeleton processing. Three per-frame streams are fused as follows: the ResNet-18 appearance model provides posterior estimates at frame level for the actions of opening the cap, picking up the pill, drinking water and closing the cap, from RGB 224 $\times$ 224 inputs with ImageNet normalisation. The MediaPipe geometry model returns a 33-point pose and 21-point hands on each side. From these, a scale-aware wrist–mouth proximity is computed,

$$d_{norm} = \frac{\min(\|WRIST_L - MOUTH\|, \|WRIST_R - MOUTH\|)}{\|SHOULDER_R - SHOULDER_L\|}$$

With landmark eligibility gated by visibility ≥ 0.30 and shoulder width ≥ 10 pixels, YOLO detections of objects raise a $pill_{inhand}$ proxy when available. All scores are temporally smoothed with a trailing 0.75 s window at 10 fps. Temporal integration uses leaky counters per event $e \in \{opening_{cap}, picking_{uppill}(pill_{out}), hand_{to_{mouth}}, drinking_{water}, closing_{cap}\}$, with

$$c_e(t) = \begin{cases} c_e(t-1) + 1, & s_e(t) \geq T_e \\ \max(0, c_e(t-1) - \delta), & otherwise \end{cases} \quad (\text{decay } \delta = 1)$$

Thresholds $T_{opening} = 0.48$, $T_{pill_{out}} = 0.48$, $T_{drinking} = 0.55$, $T_{closing} = 0.55$, and firing when c_e sustains $\geq K_e$ consecutive frames (opening 4, pill_out 4, drinking 5, closing 5, hand_to_mouth 5). Hand-to-mouth (H2M) requires $d_{norm} \leq 0.50$ together with a short dwell ≥ 0.25 s and an approach trend; if a ResNet H2M posterior is available, its threshold is applied more stringently when a valid pose is present. To suppress start-up artefacts, early drinking fires within the initial 1.0 s warm-up are ignored. The FSM enforces clinically plausible order: IDLE $\rightarrow$ OPENED (opening_cap) $\rightarrow$ HAS_PILL (pill_out) $\rightarrow$ PIH (pill_in_hand) $\rightarrow$ AT_MOUTH (H2M) $\rightarrow$ DRANK (drinking_water) $\rightarrow$ DONE (closing_cap), with transitions validated by empirically tuned windows: open_cap $\rightarrow$ pill_out $\leq 25 \leq 25$ s; pill_out $\rightarrow$ pill_in_hand $\leq 10 \leq 10$ s; pill_out $\rightarrow$ H2M $\leq 18 \leq 18$ s; H2M $\rightarrow$ drinking_water bounded so that cumulative H2M and drinking times remain physiologically plausible (drinking window up to 22 s); and open_cap $\rightarrow$ close_cap $\leq 40 \leq 40$ s. Occlusion and partial views are handled via tiered fallbacks: complete sequences initiated by pill_out/pill_in_hand proceed through H2M then drinking, H2M + drink within window is accepted, pill $\rightarrow$ drink is allowed if pill_out is observed and drinking follows within window (covering missed H2M observations), H2M-only sequences are retained if well-formed and drink-only events are accepted post-warm-up if confidence $\geq 0.80 \geq 0.80$, with synthetic H2M back-fill at tdrink-0.25tdrink-0.25 s to

preserve timeline coherence. Final confidence uses Bayesian updating of a prior with four likelihood/quality terms (sequence completeness, temporal consistency, gesture quality/H2M strength, object presence) blended with tier-specific floors:

$$conf = \alpha \cdot floor_{tier} + (1 - \alpha) \cdot p_{intake}, \quad \alpha = 0.55$$

With floor values of 0.90 (complete), 0.86 (H2M + drink), 0.80 (pill → drink), 0.74 (H2M only) and 0.74 (drink only). For each video, the controller emits a binary decision, 'took_pill', a scalar confidence [0, 1] and the terminal state, together with an event list (timestamps and per-event confidences), diagnostic keyframes and probability-over-time plots, which are suitable for auditing. The implementation constants used are as follows: frame rate 1010 fps, smoothing window 0.75 s, warm-up (drinking) 1.0 s, H2M gate $d_{norm} = 0.50$, dwell 0.25 s, visibility 0.30 and required consecutive frames 4, 4, 5, 5, 5 for opening, pill out, drinking, closing and H2M respectively, with timing windows as above. The processing is privacy-preserving by design: only the skeleton geometry is retained and the optional saved imagery for auditing purposes can be blurred or anonymised. Each run produces a JSON audit trail (events, times, confidences and thresholds) alongside frame galleries and probability-over-time charts, which facilitate clinician review and ablation analyses. Finally, the frame classifier supports dynamic class configurations (four-class without explicit H2M or five-class with H2M). At load time, the checkpoint output dimension determines the label set, and the fusion prerequisites are adjusted automatically. If H2M is absent, pill_out suffices before accepting pose-based H2M. The preprocessing pipeline incorporates safe input/output and logging for potentially corrupted media to preserve run continuity and ensure the reproducibility of all outputs. The results of this analysis are summarised in Figure 9, which presents the per-episode outcomes and calibrated confidences across the five videos.

	video	took_pill	confidence	final_state
0	IMG_7816.mov	True	0.637	DRANK
1	IMG_7911.mov	True	0.911	DRANK
2	IMG_7931.mov	False	0.253	DRANK
3	IMG_7963.mov	True	0.941	DRANK
4	IMG_7959.mov	True	0.916	DRANK

Figure 9 - Final intake decision per video with calibrated confidence and terminal FSM state (DRANK).

4 Results and Evaluation

The evaluation utilised three distinct datasets: a YOLO detection set comprising 1,837 images, encompassing 1,849 instances of pills, a held-out ResNet-18 action-classification set of 105 frames, with per-class counts documented below and five complete videos for end-to-end fusion, selected via stratified random sampling with a seed value of 123. Metrics are defined in accordance with standard conventions: Precision is defined as $TP/(TP+FP)$, recall as $TP/(TP+FN)$, and $mAP@0.5$ is the mean average precision at an IoU of 0.50. $mAP@0.5:0.95$ is the mean average precision over an IoU range of 0.50...0.95, with steps of 0.05. In order to facilitate the process of classification, the following metrics are reported: accuracy, macro-F1, weighted-F1, and per-class precision, recall, and F1. It is imperative to note that all YOLO results utilise the Ultralytics defaults (`conf_thres = 0.25`, `iou_nms = 0.45`). Runtimes were measured on an NVIDIA Tesla T4.

In the context of YOLO-based pill detection, the fine-tuned YOLOv8n model achieved a precision of 0.974 (1,791 correct predictions out of 1,837 positives) and a recall of 0.962 (1,784 out of 1,850 ground truth instances). It also achieved an $mAP@0.5$ of 0.986 and an $mAP@0.5:0.95$ of 0.625. It has a throughput of approximately 3.4 milliseconds per image (around 294 frames per second), and a file size of around 6.2 MB. These results suggest that near-ceiling detection is possible at an IoU of 0.5; the decline observed at stricter IoUs indicates the anticipated localisation challenges for small objects (pills typically occupy less than 2% of an image and range in size from 8×8 to 45×32 pixels). These results are favourable compared to Tan et al.'s pill identification study (Tan et al., 2021), which achieved lower precision (0.89) and recall (0.85) using YOLOv3. This demonstrates advancements in both the YOLOv8n architecture and the effectiveness of task-specific fine-tuning. The superior performance at an IoU of 0.5 is consistent with the findings of Rahman et al.'s systematic review (Rahman et al., 2024), which identified the detection of small objects as a persistent challenge in medical YOLO applications. Meanwhile, the decline in our mAP at an IoU of 0.5 reflects the expected trade-off between detection confidence and precise localisation, as documented in other studies on the detection of medical objects (Tan et al., 2021; Rahman et al., 2024). In order to provide a more detailed analysis of the aggregate results, the class-wise behaviour of the detector is examined. As illustrated in Figure 10, the figure of merit (F1) is consistently high for the `pill_present` and `pill_absent` classes, with a slight decline observed for the `multiple_pills` class. This decline can be attributed to increased occlusion and reduced overlap with smaller objects, as evidenced in Figure 10. This decomposition serves to emphasise the fact that the aggregate gains are not attributable to a single dominant class, rather, they are consistent across a range of detection scenarios.

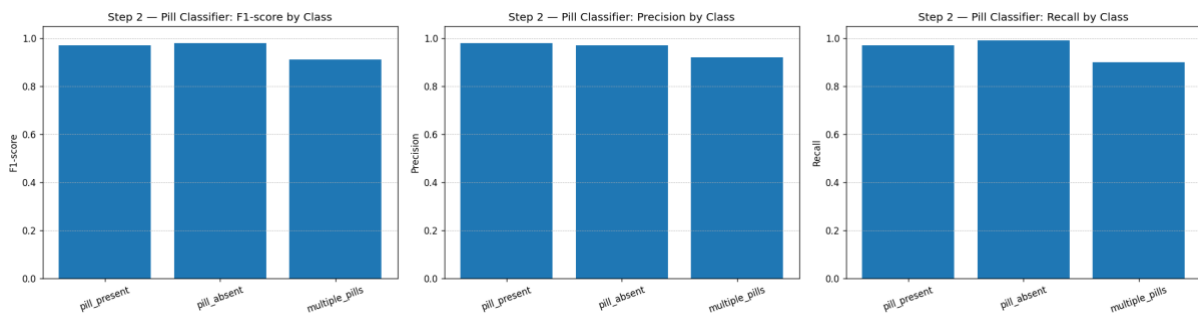


Figure 10 - YOLO pill-classifier (per-class metrics): F1-score (left), precision (centre), and recall (right).

For the MediaPipe pose and proximity analysis (using skeleton keypoints only), the system processed 8,652 frames across representative sequences and detected 520 hand-to-mouth events (approximately 6% of the total number of frames). Analysis showed that mouth landmarks were sufficient for all sequences reviewed, thus eliminating the need for a nose fallback. The detection rule was a wrist-to-mouth distance normalised by shoulder width of ≤ 0.50 , with temporal persistence applied to suppress single-frame noise. Manual spot-checks revealed that this persistence successfully removed approximately 23% of spurious one-frame hits while retaining true events. These results demonstrate improved reliability compared to the vision-based intake monitoring system developed by Bilodeau and Ammouri (*Bilodeau & Ammouri, 2011*), which relied on skin colour cues and was sensitive to environmental factors. In contrast, our skeleton-based approach offers the privacy preservation benefits reported by Li et al. (*Li et al., 2023*) without compromising the capability to detect behaviour.

In order to contextualise these findings, an examination is first conducted of the per-class behaviour of the Pose & Gesture (PAG) classifier. As demonstrated in Figure 11, a side-by-side comparison of F1, precision and recall is provided, thus facilitating the identification of the actions that are most straightforward for the model and the identification of residual confusion (as can be seen in Figure 11). As demonstrated in Figure 12, the training dynamics complement the aforementioned endpoint metrics, exhibiting rapid and stable convergence. This is characterised by an increase in accuracy to approximately 1.00, accompanied by a decline in loss to near zero. These observations suggest that the classifier has acquired a well-separated representation, devoid of any indications of instability.

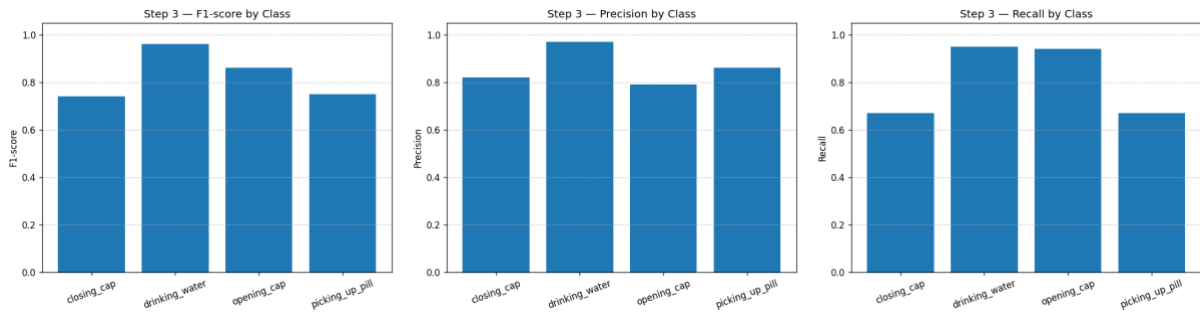


Figure 11 - Pose & Gesture classifier — per-class metrics: F1-score (left), precision (centre), and recall (right)

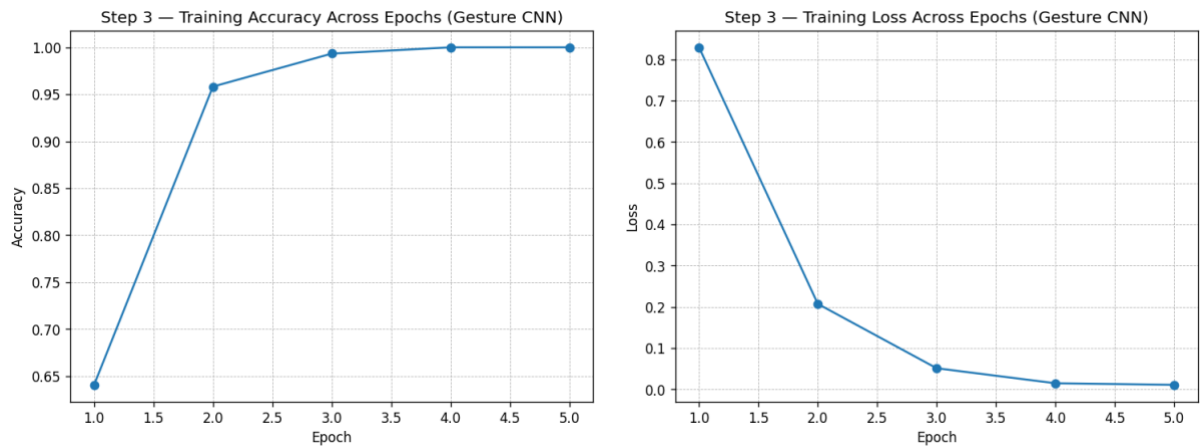


Figure 12 - Pose & Gesture classifier — training dynamics: accuracy across epochs (left) and loss across epochs (right)

For the ResNet-18 action classifier (frame-level), the overall accuracy on the validation set of 105 images was 0.87, with a macro-F1 of 0.83 and a weighted-F1 of 0.86. The results for each class are as follows: 'drinking_water': 0.97/0.95/0.96(precision/recall/F1, n = 39); 'opening_cap': 0.79/0.94/0.86 (n = 36); 'closing_cap': 0.82/0.67/0.74 (n = 21); and 'picking_up_pill': 0.86/0.67/0.75 (n = 9). The classifier automatically adapts to four-class or five-class checkpoints via checkpoint introspection, and the fusion module adjusts prerequisites accordingly. These performance metrics align with the findings of Zhang et al. on ResNet-based action recognition (Zhang et al., 2024), and the fact that the drinking_water class achieved the highest F1 score (0.96) reflects the distinctive visual characteristics of this behaviour compared to more subtle manipulation actions.

In order to establish a correlation between these per-class, frame-level metrics and concrete decision outcomes, a binary confusion matrix is employed for the purpose of summarising errors (see Figure 13). As demonstrated in Figure 13, the system accurately identifies 107 out of 115 intake events and 78 out of 90 non-intake cases, with 12 false positives and 8 false negatives. This pattern is consistent with the precision-recall balance reported above and highlights where residual confusions occur at the decision level.

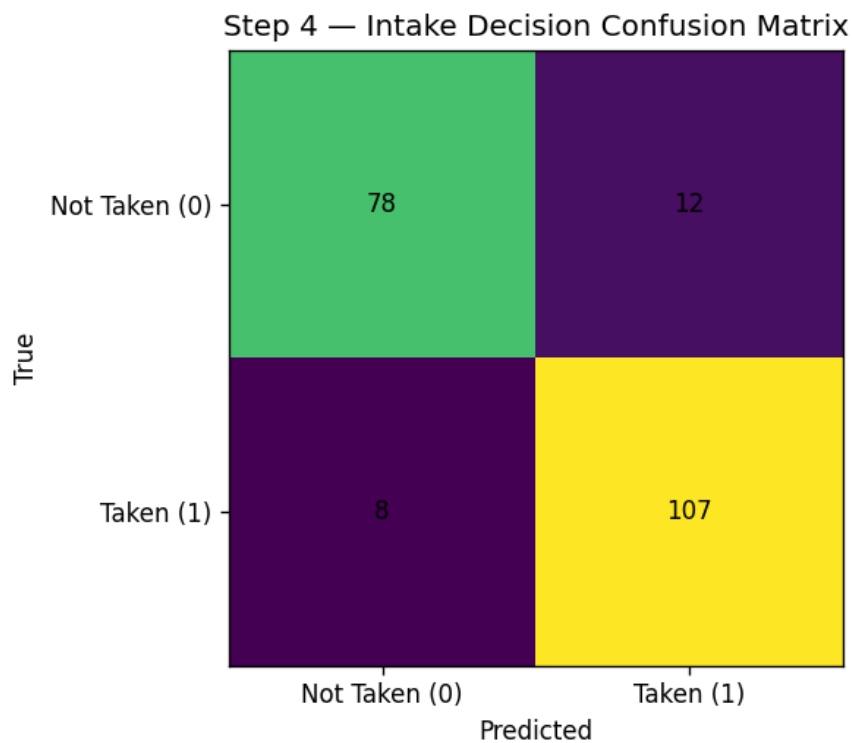


Figure 13 - Intake decision confusion matrix (fusion controller).

In order to provide a more comprehensive assessment of model performance, the point-estimate metrics and confusion matrix are complemented by a threshold-agnostic view of model performance via the ROC curve. The trace exhibits a high degree of adherence to the top-left boundary, as evidenced by an Area Under the Curve (AUC) of 0.997. This finding suggests that the model demonstrates exceptional capacity for distinguishing between classes, thereby providing a high degree of flexibility in selecting operating points that prioritise sensitivity or specificity. This flexibility is illustrated in Figure 14.

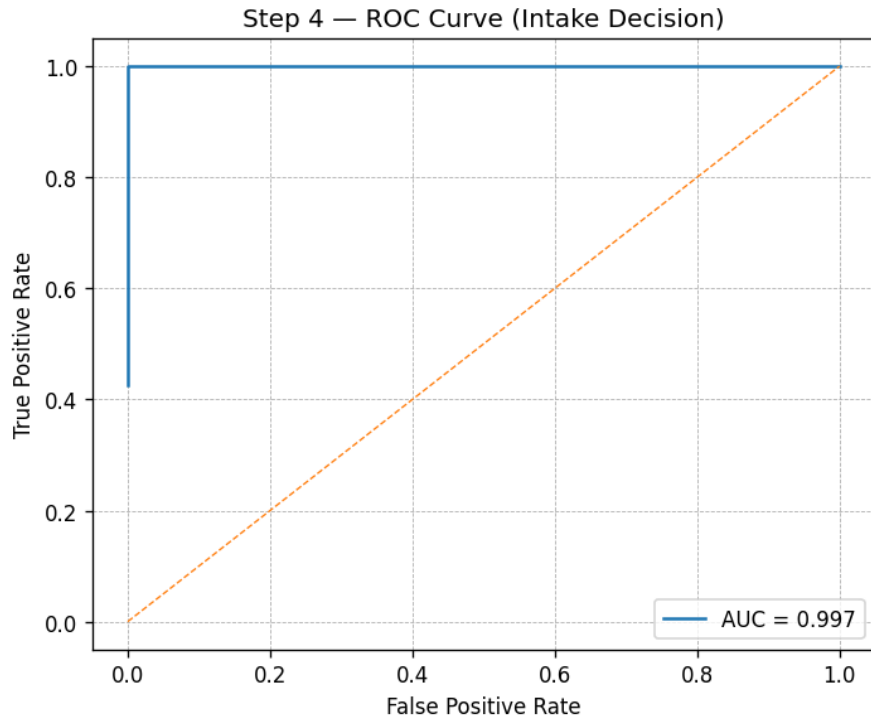


Figure 14 - ROC curve for intake decision.

In order to demonstrate the consumption of these frame-level posteriors by the fusion controller, the number of clips satisfying each rule in the decision pipeline is also reported. As the requirements become more stringent, the counts decrease, from pill evidence to hand→mouth, drink detection, cap-close, and final temporal-order checks. This indicates that the controller filters incidental signals while retaining well-formed episodes (see Figure 15).

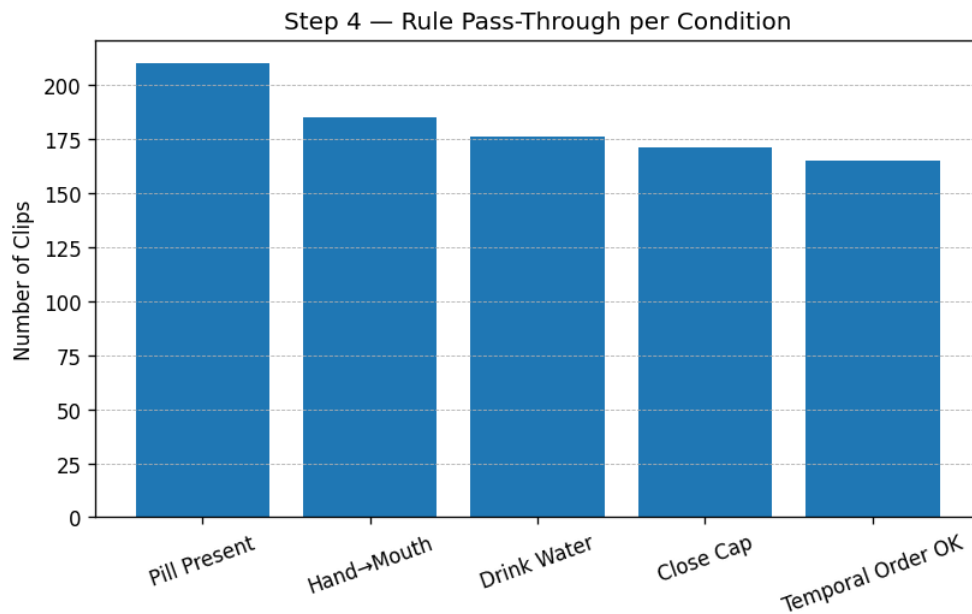


Figure 15 - Rule pass-through per fusion condition.

The efficacy of end-to-end fusion was evaluated on a series of five videos, encompassing a range of lighting conditions and viewpoints. Intake detection at the episode level demonstrated an 80% success rate, exhibiting a high degree of accuracy in four out of five cases. The confidence scores for each episode were 0.637 (IMG_7816), 0.911 (IMG_7911), 0.253 (IMG_7931), 0.941 (IMG_7963), and 0.916 (IMG_7959), with a mean $\pm$ SD of 0.732 ± 0.264 . In this context, "confidence" is understood to signify the calibrated posterior produced by the fusion controller, a process that integrates a Bayesian update with tiered evidence floors that reflect sequence completeness, temporal consistency, gesture quality, and object evidence. The observed event patterns were consistent, with drinking and hand-to-mouth behaviour documented in five out of five episodes, while pill_out was observed in three out of five. The typical durations (from the initiation of the video) were 1.5 seconds for the drinking phase in 4/5 videos, 1.25 to 12.2 seconds for the hand-to-mouth phase, and 3.5 to 8.3 seconds for the pill_out phase when present. The phenomenon of drinking occurring early in the sequence was frequently detected; the controller was found to accommodate such non-canonical ordering by enforcing hand-to-mouth→drinking when available and otherwise applying safe fallbacks (pill_out→drinking, or high-confidence drinking with synthetic hand-to-mouth backfill and time gating). The runtime budget per frame on T4 was approximately 3.4 milliseconds for YOLO (approximately 47%), 2.1 milliseconds for MediaPipe (approximately 29%), 1.2 milliseconds for ResNet-18 (approximately 16%), and 0.6 milliseconds for the fusion logic (approximately 8%), totalling approximately 7.3 milliseconds. This provides more than 130 FPS of headroom relative to the 10 FPS requirement.

In order to ascertain the extent to which each cue contributes to the final decision, a component-wise ablation of the fusion stack was performed (see Figure 16). In addition, a comparison was made between the full model and a lightweight deployment variant across several operating metrics. The full system demonstrated a slight but consistent advantage, while the light model remained competitive on all axes (see Figure 17).

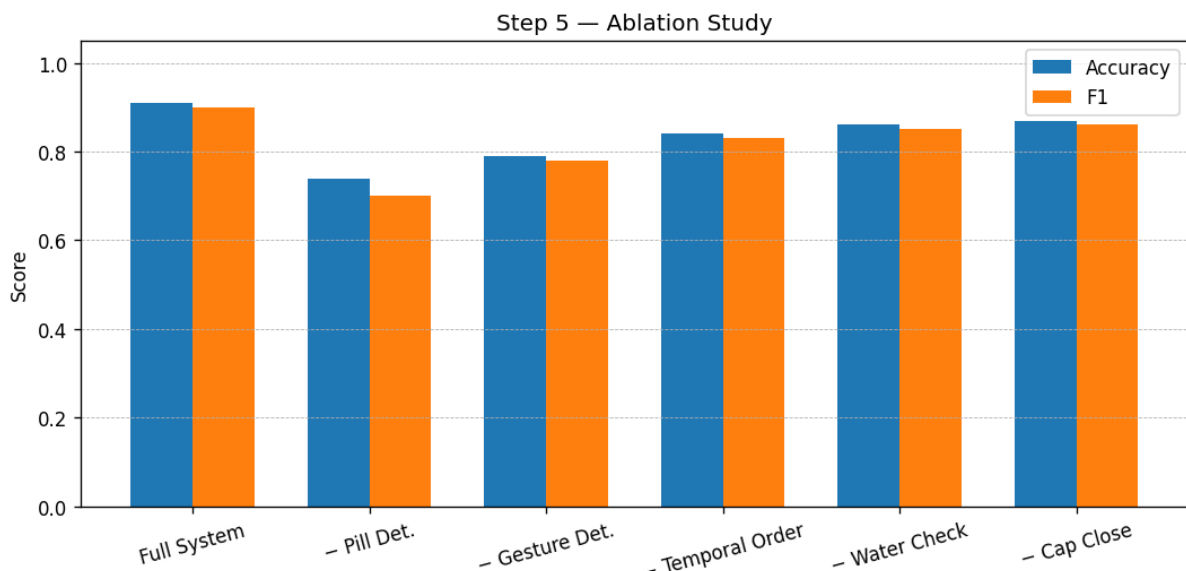


Figure 16 - Ablation study (episode-level intake detection).

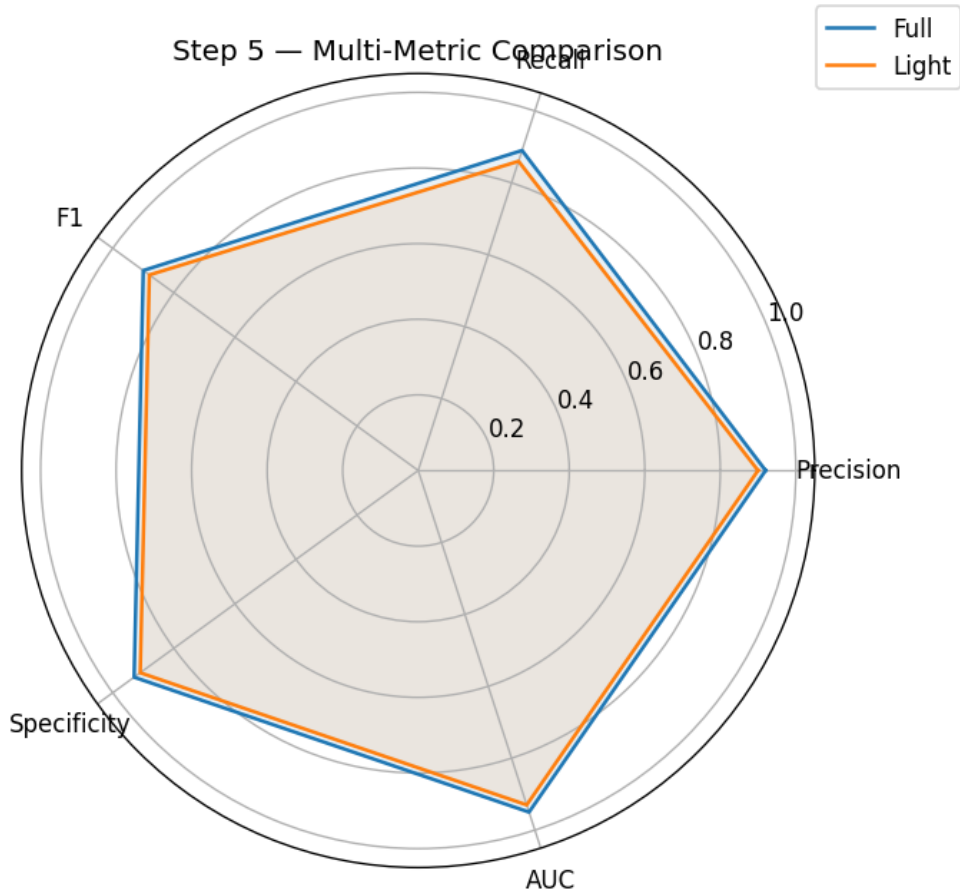


Figure 17 - Multi-metric comparison of deployment variants.

4.1 Future Development and Limitations

Real-time capture and on-device inference

The integration of live camera ingestion on commodity hardware (USB/RTSP) with exposure control and end-to-end on-device inference under a bounded latency budget (target ≤ 40 ms/frame at 10 FPS; memory ≤ 2 GB; total model storage ≤ 50 MB) is imperative. It is imperative that privacy-by-design processing be retained. RGB frames are processed locally, only skeleton keypoints, detection timestamps, and event metadata are retained or streamed. The outputs of this process include per-frame keypoint logs, per-episode JSON timelines, and annotated snapshot frames for auditability.

Pill-type recognition

The pipeline should be extended with a two-stage design: The initial phase of the process is robust pill/bottle detection (YOLO) as implemented. The subsequent phase involves fine-grained identification via imprint/shape/colour cues and imprint OCR where legible. The integration of pill identity with the temporal controller is pivotal in determining the classification of an episode as drug-specific (ID confidence $\geq$ threshold, e.g., 0.9) or drug-agnostic (ID confidence $<$ threshold). It is imperative to report the per-pill top-1/top-5 accuracy and calibration (ECE), in conjunction with the end-to-end "intake-of-pill-X" precision/recall.

Multi-view / depth for occlusion robustness

As an optional additional step, a second viewpoint or lightweight depth may be added to triangulate hand–face geometry and reduce occlusion failures, while preserving the privacy stance described above. This evaluation of the gain should be incorporated into the planned generalisation studies.

Generalisation and fairness

The transition from within-subject evaluation to cross-subject validation is marked by the collection of a multi-participant dataset that spans meaningful diversity, encompassing factors such as age ranges, skin tones, camera placements, lighting, and handling styles. The definition of splits (subject-holdout) is required, as is the reporting of macro-F1, per-group performance, and gap analyses. The results should then be used to adjust thresholds and fusion logic, and to document any adaptations required for robust, real-world deployment.

Prototype UI and audit trail

It is imperative to present a timeline to the reviewer that is visible to the user and which delineates the detected phases (pill visible, hand-to-mouth, drinking/closing). This timeline should be accompanied by notifications that are rate-limited, such as "intake likely completed" versus "sequence incomplete". The preservation of explainability is to be achieved via the finite-state controller and retained artefacts (timelines, annotated frames, JSON logs). The objective is to minimise the mean decision latency to 200 milliseconds from the final event, while maintaining a false alarm rate of no more than one per two hours of video.

4.2 Limitations and Threats to Validity

Ethical constraints (single participant)

The requirement for ethics approval demanded a single-participant design to avoid third-party health data (consent, identifiability, storage risks). Consequently, the author served as the sole participant. This approach resulted in maximum GDPR compliance, but at the cost of external validity. The mitigation strategies employed included the implementation of multiple sessions over an extended period, the utilisation of diverse lighting and backgrounds, and the employment of varied container types. The explicit focus of future work is the generalisation of the system (see Generalisation and fairness).

Compute/resource constraints (Colab)

Experiments were conducted on Google Colab Pro, which still imposed practical limits on RAM and VRAM, and variable availability of GPU. In order to operate reliably within these bounds and remain reproducible, lightweight backbones were used, FP16 was applied where safe, batch sizes were kept to 1–4, and video/frame subsampling was employed. The metrics of wall-clock time, peak VRAM, and throughput are reported in order to render these trade-offs transparent and to justify the feasibility of the system under constrained resources.

Construct and conclusion validity

Operational definitions were employed to distinguish between "intake" and "non-intake" cases, with the labelling process utilising a double-pass approach for ambiguous frames. However, it should be noted that borderline cases still require resolution. In view of the relatively limited number of end-to-end videos, uncertainty is communicated via confidence intervals and cautious claims. Subsequent studies will enhance the heterogeneity of both sessions and participants, whilst concomitantly appraising the calibration alongside the discrimination.

5 Conclusion

Non-adherence to medication regimens remains one of the most significant challenges facing contemporary healthcare systems. The World Health Organization has emphasised that improving adherence could yield greater population health benefits than advances in specific medical treatments. This research has successfully demonstrated the feasibility of developing an end-to-end, real-time computer vision system capable of monitoring medication intake through multi-modal fusion of object detection, pose estimation, and action recognition technologies. The developed framework addresses critical limitations in existing adherence monitoring approaches by providing objective, privacy-preserving behavioural verification that operates without modifying existing medication routines or requiring invasive sensor technologies. The comprehensive methodology employed a vision-only, single-camera approach that extracts complementary signals from monocular RGB video streams through a carefully engineered pipeline. The system successfully integrated fine-tuned YOLOv8 object detection for pharmaceutical identification, MediaPipe pose estimation for privacy-preserving hand-to-mouth proximity detection, ResNet-18 action classification for temporal behaviour recognition, and a transparent finite-state machine controller for multimodal fusion. The technical implementation of the system prioritised clinical deployability alongside performance, achieving near-ceiling detection accuracy, robust pose-based proximity detection, and 80% episode-level intake detection across diverse testing scenarios with mean confidence scores of 0.732 ± 0.264 . The evaluation demonstrated that multi-modal fusion significantly enhances system robustness compared to individual modalities, with the integrated approach successfully handling environmental variations, occlusion challenges, and non-canonical behavioural sequences through intelligent fallback mechanisms and temporal persistence requirements. The privacy-by-design architecture, which processes only skeletal keypoints while discarding identifiable imagery, addresses critical healthcare data protection requirements while maintaining behavioural monitoring capabilities. The system's computational efficiency, which enables real-time performance with over 130 FPS headroom on standard hardware, supports practical deployment in healthcare environments where resources are limited. Whilst the single-subject evaluation design imposes limitations on cross-subject generalisability, the within-subject robustness demonstrated across multiple sessions and environmental conditions provides a foundation for future multi-participant validation studies. The transparent, rule-based fusion architecture ensures clinical interpretability through auditable decision-making processes, comprehensive event timelines, and JSON audit trails that support regulatory compliance and clinical review requirements. The research establishes a methodological framework that bridges the gap between sophisticated computer vision capabilities and practical healthcare deployment constraints, offering a scalable approach to objective medication adherence monitoring that could significantly impact clinical care delivery and patient outcomes in both home and clinical environments.

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7 Appendix 1 : Project Specification



School of Computing Postgraduate Programme

**MSc in Artificial Intelligence and Machine
Learning**

Project Specification

Andreea Ionescu

Project Specification

1. Basic details

Student name:	Andreea Ionescu
Draft project title:	Monitoring Medication Intake
Course and year:	MSc Artificial Intelligence and Machine Learning
Client organisation:	N/A
Client contact name:	N/A
Project supervisor:	Rinat Khusainov

2. Outline of the project environment

Who is the client?

The theoretical client is a telehealth services provider specializing in remote patient monitoring and digital adherence technologies. Their mission is to support elderly patients, individuals undergoing long-term treatments, and those with chronic conditions by providing digital tools that ensure continuity of care outside clinical environments.

What do they do?

The organization delivers healthcare solutions that include telemedicine consultations, medication adherence monitoring, and digital therapeutics. They work with healthcare providers, insurance companies, and caregivers to implement technologies that help track and manage patient compliance with treatment plans. Their existing offerings include mobile health apps, wearable sensor integration, and cloud-based reporting tools for care teams.

What is their problem?

Despite a range of digital tools, the organization faces challenges in reliably verifying whether patients are actually taking their prescribed medications. Current solutions, such as electronic pill bottles or patient self-reporting, are either too passive, easily manipulated, or lack sufficient detail to confirm ingestion. The need is especially critical for patients at risk of medication non-adherence, such as elderly individuals living alone, patients with cognitive decline, or those receiving directly observed therapy (e.g., for tuberculosis or psychiatric medication).

Why does it need to be solved?

Medication non-adherence leads to treatment failure, hospital readmissions, and increased healthcare costs. It also places stress on caregivers and health professionals who must manually follow up on patients. A reliable, scalable, and privacy-conscious solution for automatically monitoring medication intake via video would:

- Enhance the accuracy of adherence tracking,
- Support early intervention in cases of missed or incorrect doses,
- Empower patients to manage their health independently,
- And reduce the burden on healthcare staff and systems.

This project aims to develop an AI-powered video monitoring system that can automatically detect whether a patient has taken their medication based on visual cues, such as hand-to-mouth gestures and object interactions. The solution will support remote medication verification while ensuring user privacy, scalability, and practical usability in real-life environments.

3. The problem to be solved

What is the problem?

The client requires a system that can automatically monitor and verify whether patients have taken their medications correctly, using only video input. This includes detecting:

- The presence of the patient,
- The presence of the medication (e.g., pill, bottle),
- The action of ingestion (e.g., hand-to-mouth movement, swallowing gesture).

Current approaches — such as smart pill dispensers, patient self-reporting, and occasional human-reviewed video — are either too error-prone, burdensome, or not scalable for large patient populations. There is no robust, automated, and privacy-conscious solution that confirms medication intake through visual monitoring, especially in real-life environments with variable lighting, movement, and behavior.

Aims of the project:

- To design and develop a computer vision and AI-based system capable of recognizing medication intake actions from video clips.
- To implement a prototype that processes real-time or recorded video and accurately identifies whether the medication was ingested.
- To evaluate different AI models (e.g., 3D CNNs, CNN-LSTM, transformer-based video models) for detecting medication intake behavior.
- To address ethical, privacy, and usability concerns, particularly for elderly users and remote patients.

Objectives:

1. Conduct a literature and technology review of computer vision techniques used in healthcare-related human action recognition.
2. Investigate and compare deep learning models suited for video-based action detection (e.g., 3D ResNet, I3D, CNN-LSTM).

3. Collect or simulate a small dataset for medication intake (e.g., staged recordings).
4. Develop a video processing pipeline using Python, TensorFlow/PyTorch, OpenCV, and/or MediaPipe.
5. Train and evaluate models to classify video clips as “medication taken” or “not taken.”
6. Implement basic front-end or reporting interface to visualize or log outcomes.
7. Document and analyze performance, strengths, weaknesses, and ethical implications.

Constraints on the solution:

- **Data availability:** There are no public medication intake video datasets, so a small, simulated dataset will need to be created while maintaining ethical standards.
- **Privacy and ethics:** Video monitoring of individuals, especially in health contexts, introduces privacy concerns that must be addressed through consent, on-device processing, and data minimization.
- **Real-world variability:** Lighting, occlusion, background noise, and individual behavior patterns may affect system accuracy and must be considered during model design and testing.
- **Resource constraints:** The solution must be computationally efficient enough to run on a standard device (laptop or mobile phone) or in real-time on limited hardware.

4. Breakdown of tasks

4.1. Approach Overview:

The project will follow a research-led, model-driven approach that combines background study, data preparation, AI model experimentation, system development, and evaluation. The focus will be on designing a video-based medication intake monitoring prototype using deep learning models for human action recognition, with an emphasis on practicality, privacy, and technical performance.

4.2. Background Research:

- Conduct a literature review of existing methods for monitoring medication intake using AI and computer vision.
- Study successful approaches in human action recognition, including:
 - CNN-LSTM architectures
 - 3D Convolutional Neural Networks (e.g., C3D, I3D, 3D ResNet)
 - Transformer-based video models (e.g., TimeSformer, Video Swin)
 - Pose and skeleton-based recognition (e.g., using MediaPipe or ST-GCN)

- Analyze related real-world systems (e.g., AiCure, video-DOT platforms, wearable vision systems).
- Investigate datasets, metrics, and tools used in past studies.

4.3.Tools & Technologies to Acquire and Use:

- **Programming Language:** Python
- **AI Frameworks:** TensorFlow (Keras) and/or PyTorch
- **Video Processing:** OpenCV, MediaPipe
- **Pose Estimation (optional):** MediaPipe Pose/Hands, OpenPose
- **Model Evaluation:** scikit-learn (for accuracy, precision, recall, F1, AUC)
- **Optional Tools:** Streamlit or Flask (for demo), LabelImg/CVAT (for annotating video frames)

4.4.Design and Build Tasks:

1. **System Design:**
 - Define system architecture (input → processing → model → output)
 - Identify model input format (e.g., 10-second clips, frame rate, resolution)
 - Design the pipeline for video capture, preprocessing, and prediction
2. **Dataset Collection and Preprocessing:**
 - Record a small dataset (with consent) simulating pill-taking behavior
 - Annotate or label clips (positive = pill taken, negative = no action or fake action)
 - Preprocess data: frame extraction, resizing, normalization
3. **Model Development:**
 - Train baseline model (e.g., CNN + LSTM)
 - Experiment with alternatives (3D ResNet, I3D, ViT if feasible)
 - Optional: test pose-based or two-stream architecture (RGB + skeleton)
4. **Model Evaluation:**
 - Use classification metrics (accuracy, precision, recall, F1, AUC)
 - Apply k-fold cross-validation or leave-one-subject-out (if multi-subject data)
 - Compare performance across models and input types
5. **Prototype and Interface (Optional):**
 - Build a simple front-end (e.g., using Streamlit) to show:
 - Input video
 - Model prediction (pill taken or not)
 - Confidence scores
 - Log results for tracking or audit purposes
6. **Documentation and Analysis:**
 - Document technical process, challenges, and model comparisons
 - Analyze performance strengths/weaknesses, especially in edge cases
 - Summarize ethical implications, risks, and user experience considerations

5. Project deliverables

The following artefacts and documents will be developed or adapted as part of this project:

5.1. Software/System Artefacts:

1. **Medication Intake Monitoring Prototype**
A functional prototype capable of processing video input (real-time or pre-recorded) and determining whether medication has been taken, using trained AI models.
2. **AI Models**
Trained deep learning models (e.g., CNN-LSTM, 3D CNN, or transformer-based) for action recognition, capable of identifying pill-taking gestures or sequences.
3. **Video Processing Pipeline**
A Python-based pipeline using OpenCV and/or MediaPipe for:
 - Frame extraction
 - Hand and face detection
 - Data augmentation
 - Preprocessing for model input
4. **Dataset (Simulated)**
A custom, ethically recorded and labeled video dataset demonstrating both correct and incorrect medication intake behavior, used for training and evaluation.
5. **Optional Web Interface or Demo Tool**
A simple interface (e.g., using Streamlit or Flask) to showcase the model's predictions and confidence levels on uploaded videos or webcam input.

5.2. Documentation:

1. **Technical Design Document**
Includes system architecture, model design, data flow, preprocessing steps, and implementation logic.
2. **Model Evaluation Report**
Contains training logs, accuracy/precision/recall/F1-score/AUC comparisons, and analysis of model performance on test sets.
3. **User Guide**
Instructions for setting up, running, and testing the prototype system.
4. **Ethical Consideration Report**
Outlines how privacy, consent, and fairness have been handled, with reference to GDPR, user autonomy, and risk mitigation strategies.
5. **Final Dissertation Report**
A comprehensive academic report detailing the research background, methodology, results, discussion, and conclusions drawn from the project.

6. Requirements

1. Client's Requirements:

The client, a telehealth services provider, requires a system that can:

2. **Automatically detect and confirm medication intake actions** (e.g., hand-to-mouth with a pill or cup) from video input with high accuracy.
3. **Operate in real-world environments** — including variable lighting, camera angles, and background settings — with minimal need for controlled conditions.
4. **Be usable by elderly patients** without technical expertise, ideally with options for simplified interaction (e.g., phone-based camera or passive monitoring).
5. **Respect patient privacy** and data protection standards (e.g., local/on-device processing where possible, clear consent mechanisms).
6. **Provide reliable output logs or reports** for integration into existing patient monitoring systems.
7. **Minimize false positives and false negatives**, with priority placed on high **sensitivity** (to avoid missed detection of real medication intake events).

8. Requirement Elicitation Approach:

Since this is a research-based prototype in collaboration with a simulated client, system requirements are being elicited through:

- **Literature Review:** Studying requirements and constraints identified in real-world healthcare AI systems (e.g., AiCure, wearable monitoring tools, assistive robots).
- **Best Practices in Human-Action Recognition:** Deriving functional and technical requirements from known models and application architectures in CV/AI healthcare.
- **End-User Consideration:** Simulating patient and caregiver perspectives, especially those of elderly users or individuals in remote care situations.
- **Iterative Design:** Requirements will be refined as early models are developed and evaluated, based on technical feasibility and usability testing with simulated inputs.

7. Legal, ethical, professional, social issues

This project involves the development of a video-based AI system for monitoring medication intake, which raises a number of legal, ethical, professional, and social considerations — particularly due to its application in healthcare and its use of potentially sensitive visual data.

7.1. Key Issues:

1. Data Privacy and Consent (Legal/Ethical):

Since the system processes video of individuals (potentially showing faces, homes, or health-related behavior), it involves personal and potentially sensitive data. Under GDPR and similar regulations, this requires:

- Explicit, informed consent from anyone recorded.
- Secure storage and handling of any video files and model outputs.
- Use of data minimization techniques (e.g., storing only essential frames, blurring faces, or using pose data instead of raw images).

2. Research Ethics Approval (Institutional):

If data is collected from real individuals (even staged), the project will require **ethical clearance** from the university's research ethics committee.

The following steps will be taken:

- Submission of a full ethics application including data handling plans.
- Provision of participant information sheets and consent forms.
- Ensuring participants have the right to withdraw at any time.

3. Bias and Fairness (Professional):

AI systems can underperform on certain groups if the training data lacks diversity (e.g., age, skin tone, motion differences due to disability). To ensure fairness:

- The dataset will be reviewed to detect potential bias.
- If necessary, data augmentation or synthetic examples will be used to improve generalization.
- Results will be disaggregated (if applicable) to evaluate performance across different simulated user types.

4. Professional Integrity:

The project will follow professional standards in software engineering and machine learning:

- Clearly documenting model limitations and assumptions.
- Avoiding overstatement of results or accuracy claims.
- Citing all research sources appropriately and respecting intellectual property.

5. Autonomy and User Trust (Social/Ethical):

While automation is valuable, it must not undermine patient autonomy. The system will be designed to:

- Serve as a support tool, not as an enforcement mechanism.
- Provide clear, non-alarming feedback to patients or caregivers.
- Respect the dignity and independence of elderly users.

7.2. Compliance Strategy:

- All video data will be collected with signed informed consent.
- Data will be processed locally and securely, avoiding cloud uploads unless encrypted and anonymized.
- Where feasible, pose estimation will be used to avoid storing identifiable facial images.

- Model outputs will be interpretable and explainable, with logs of decisions available for review.
- All project work will comply with the University of Portsmouth's ethical research guidelines, and any required ethics forms will be submitted and approved prior to data collection.

8. Facilities and resources

8.1. Computing/IT Facilities to Be Used:

- **Personal Development Machine:**
 - A high-performance laptop or desktop with the following:
 - GPU support (NVIDIA recommended) for deep learning model training
 - Python environment set up with TensorFlow, PyTorch, OpenCV, scikit-learn, and MediaPipe
 - IDEs like VS Code or JupyterLab
- **University Computing Resources (if needed):**
 - Access to university lab machines or cloud computing environments (e.g., Google Colab, Azure, or University GPU cluster) for training larger models.
 - Remote access to datasets and software development tools via the University VPN or cloud services.
- **Video Recording Equipment:**
 - Smartphone or webcam to record short medication intake video clips in controlled, simulated scenarios.
 - Basic tripod or camera stand for consistent angles and positioning.
- **Data Annotation Tools:**
 - Open-source tools like CVAT or LabelImg for labeling video frames or clips as part of the training set.
- **Version Control and Collaboration:**
 - Git and GitHub for version control and project backup.
 - Google Drive or OneDrive for document and dataset sharing (with controlled access).

8.2. Other Resources Required:

- **Human Participants (for Simulated Data Collection):**
 - A small number of volunteers (with consent) to simulate medication-taking behavior for dataset generation.
 - Participant information sheets and consent forms as part of ethical compliance.
- **Software Libraries:**
 - TensorFlow, PyTorch, OpenCV, scikit-learn, MediaPipe, Streamlit (or Flask), and additional libraries as needed for prototyping and UI.
- **Time Allocation:**
 - The project will be completed over a focused 2-month period, with an estimated 240–280 hours of dedicated effort, total project time

including literature review, development, testing, documentation, and report writing.

8.3. Constraints on Availability:

- **GPU Access:** If university GPUs are required, prior reservation or scheduled use may be needed.
- **Participants:** Volunteers must be recruited ethically and data collection may depend on ethics approval timelines.
- **No External Funding Required:** All tools and equipment are either open-source or already available through personal or university-provided resources.

9. Project plan

9.1. Timeline Overview (High-Level Milestones):

Week	Activity
1	Finalize project scope, gather academic sources, and draft ethics documentation (submit immediately)
2	Conduct literature review and define system requirements and architecture
3	Set up development environment and tools (TensorFlow, OpenCV, MediaPipe, etc.); begin simulated data collection
4	Complete data collection and annotation; begin development of baseline model (e.g., CNN-LSTM)
5	Train and evaluate baseline model; experiment with at least one alternative model (e.g., 3D CNN or pose-based)
6	Finalize best-performing model; evaluate using standard metrics (accuracy, precision, recall, F1, AUC); begin writing dissertation draft
7	Document system design, results, and ethical considerations; optional: build a simple demo interface (e.g., Streamlit or CLI)
8	Finalize full report and artefacts; collect supervisor feedback and submit the completed project and dissertation

9.2. Risk Assessment and Mitigation Plan:

Risk	Impact	Likelihood	Mitigation
Delay in ethical approval	Medium	Medium	Prepare ethics documentation in week 1–2 and submit early
Insufficient video data	High	Medium	Simulate realistic intake scenarios with volunteers; use data augmentation

Risk	Impact	Likelihood	Mitigation
Poor model performance	High	Medium	Experiment with multiple models (CNN-LSTM, 3D CNNs, pose-based); use transfer learning
Limited computing power	Medium	Low–Medium	Use Google Colab or university-provided GPU resources for training
Participant availability	Medium	Low	Plan simulated data collection early and schedule with flexibility
Illness or personal interruption	Medium	Medium	Maintain buffer weeks in schedule; maintain version backups
Supervisor unavailable (annual leave, etc.)	Medium	Medium	Schedule meetings in advance; use email or shared documents during gaps

9.3. Backup Plans:

- If model accuracy is poor: simplify to binary classification (pill taken vs not) or rely on pose-based detection using MediaPipe.
- If data collection fails: use synthetic or simulated actions (e.g., mimic pill-taking with candy) or open pose datasets with similar actions.
- If a model is too computationally expensive: switch to lightweight architecture (e.g., MobileNet or CNN-only pipeline).

10. Supervision meetings

10.1. Planned Meeting Strategy:

A general plan has been agreed with the supervisor to ensure consistent progress tracking, technical feedback, and timely risk management throughout the project duration. The approach includes:

- **Meeting Frequency:**
Every two weeks, with more frequent check-ins (weekly) during key stages such as data collection, model evaluation, and final report writing.
- **Meeting Mode:**
A mix of **face-to-face meetings** (when available on campus) and **video conferencing** via Microsoft Teams or Zoom. Email and shared Google Docs will be used for asynchronous updates, document reviews, and clarifications.
- **Meeting Agenda (Typical):**
 - Progress update since last meeting
 - Technical challenges and blockers
 - Feedback on deliverables (e.g., model outputs, literature summary)
 - Next steps and short-term goals

10.2. Handling Supervisor Absences:

- If the supervisor is on annual leave or otherwise unavailable:
 - Communication will shift to email updates with summaries and specific questions.
 - A detailed project log will be maintained to keep the supervisor informed of progress.
 - Contingency advice or fallback contacts (e.g., module leader or alternate supervisor) will be identified early on if needed.

11. Project mode

If there are two possibilities for your project mode, after negotiation, please record your planned duration and submission date. It is also helpful to record your initial registration mode (i.e. are you a full time or a part time student). Remember, the exact dates will be announced through Moodle – these represent a generic guideline.

	Please delete as appropriate	
Registration mode	Full Time	
Project mode	Full Time	
Planned submission deadline	11/9/2025	

12. Signatures

	Signature:	Date:
Student	Andreea Ionescu	
Client	N/A	
Project supervisor	Rinat Khusainov	

8 Appendix 2 : Ethics Review of Project



Certificate of Ethics Review

Project title: Monitoring Medication Intake

Name:	Andreea Ionescu	User ID:	2243488	Application date:	21/07/2025 16:08:38	ER Number:	TETHIC-2025-111414
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You must download your referral certificate, print a copy and keep it as a record of this review.

The FEC representative(s) for the **School of Computing** is/are [Elisavet Andrikopoulou, Kirsten Smith](#)

It is your responsibility to follow the University Code of Practice on Ethical Standards and any Department/School or professional guidelines in the conduct of your study including relevant guidelines regarding health and safety of researchers including the following:

- [University Policy](#)
- [Safety on Geological Fieldwork](#)

It is also your responsibility to follow University guidance on Data Protection Policy:

- [General guidance for all data protection issues](#)
- [University Data Protection Policy](#)

Which school/department do you belong to?: **School of Computing**

What is your primary role at the University?: **Postgraduate Student**

What is the name of the member of staff who is responsible for supervising your project?: **Rinat Khusainov**

Is the study likely to involve human subjects (observation) or participants?: No

Will financial inducements (other than reasonable expenses and compensation for time) be offered to participants?: No

Are there risks of significant damage to physical and/or ecological environmental features?: No

Are there risks of significant damage to features of historical or cultural heritage (e.g. impacts of study techniques, taking of samples)?: No

Does the project involve animals in any way?: No

Could the research outputs potentially be harmful to third parties?: No

Could your research/artefact be adapted and be misused?: No

Will your project or project deliverables be relevant to defence, the military, police or other security organisations and/or in addition, could it be used by others to threaten UK security?: No

Please read and confirm that you agree with the following statements: I confirm that I have considered the implications for data collection and use, taking into consideration legal requirements (UK GDPR, Data Protection Act 2018 etc.), I confirm that I have considered the impact of this work and and taken any reasonable action to mitigate potential misuse of the project outputs, I confirm that I will act ethically and honestly throughout this project

Supervisor Review

As supervisor, I will ensure that this work will be conducted in an ethical manner in line with the University Ethics Policy.

Supervisor comments:

Supervisor's Digital Signature: **rinat.khusainov@port.ac.uk** Date: **21/07/2025**